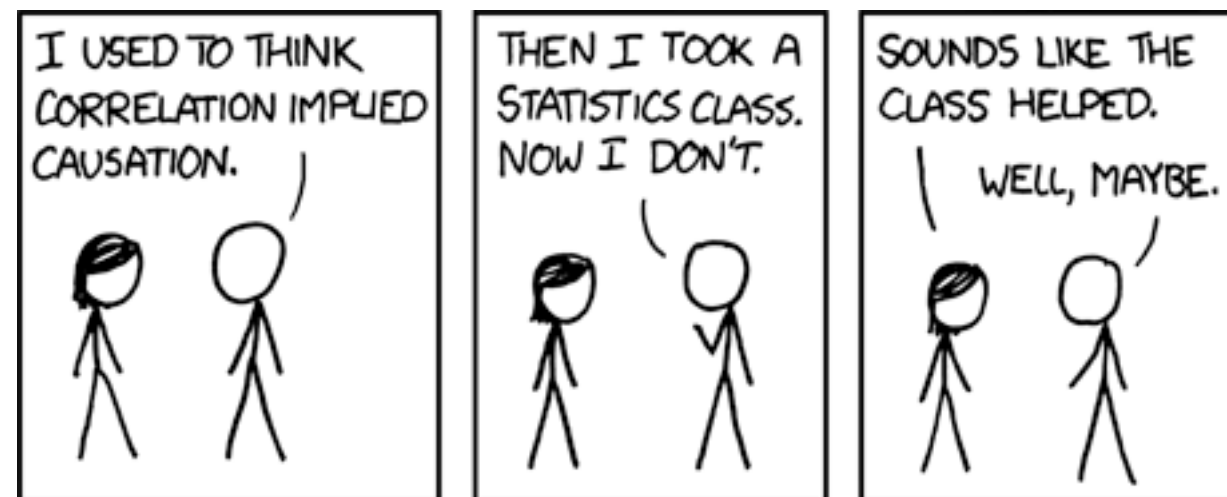


Introduction to Causal Inference with Machine Learning

Armeen Taeb & Ali Shojaie

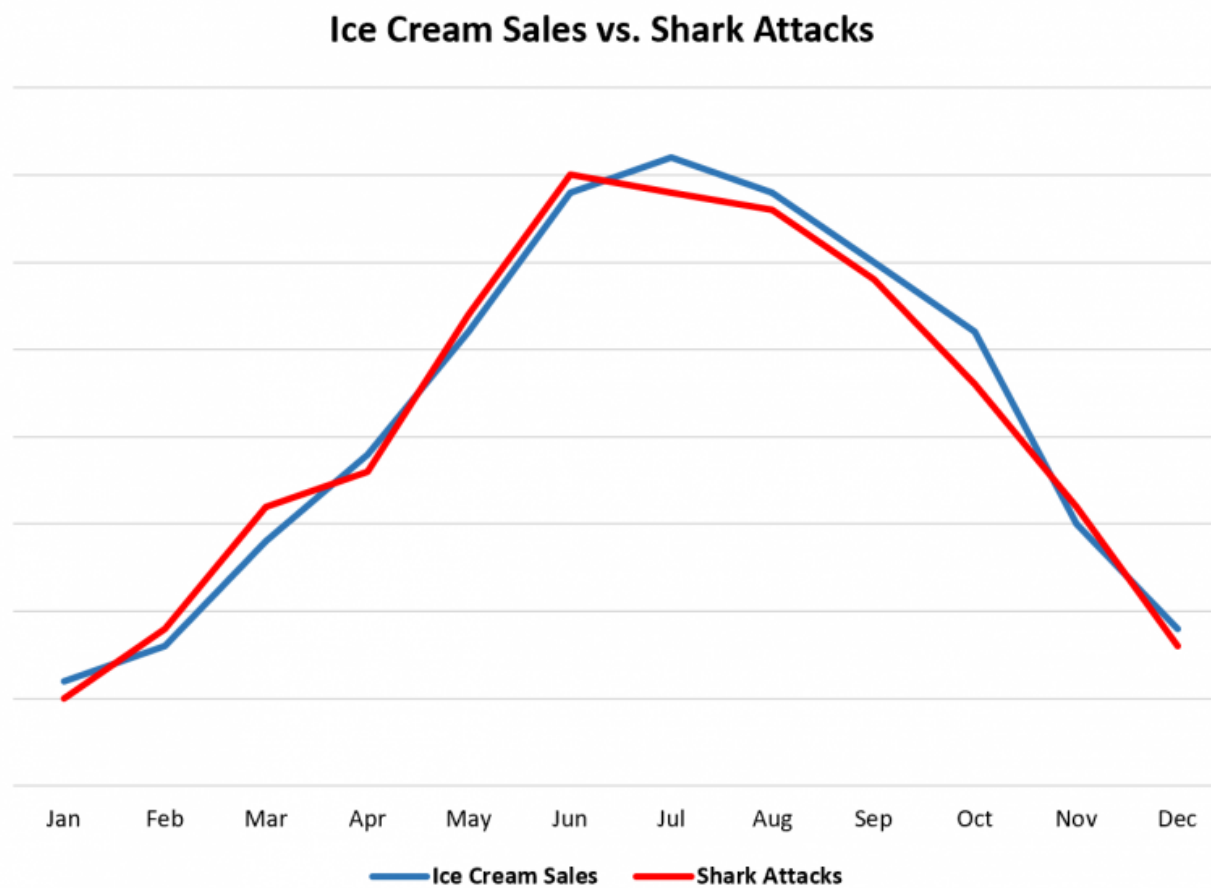
July 22-24, 2026

Summer Institute in Statistics for Big Data
University of Washington



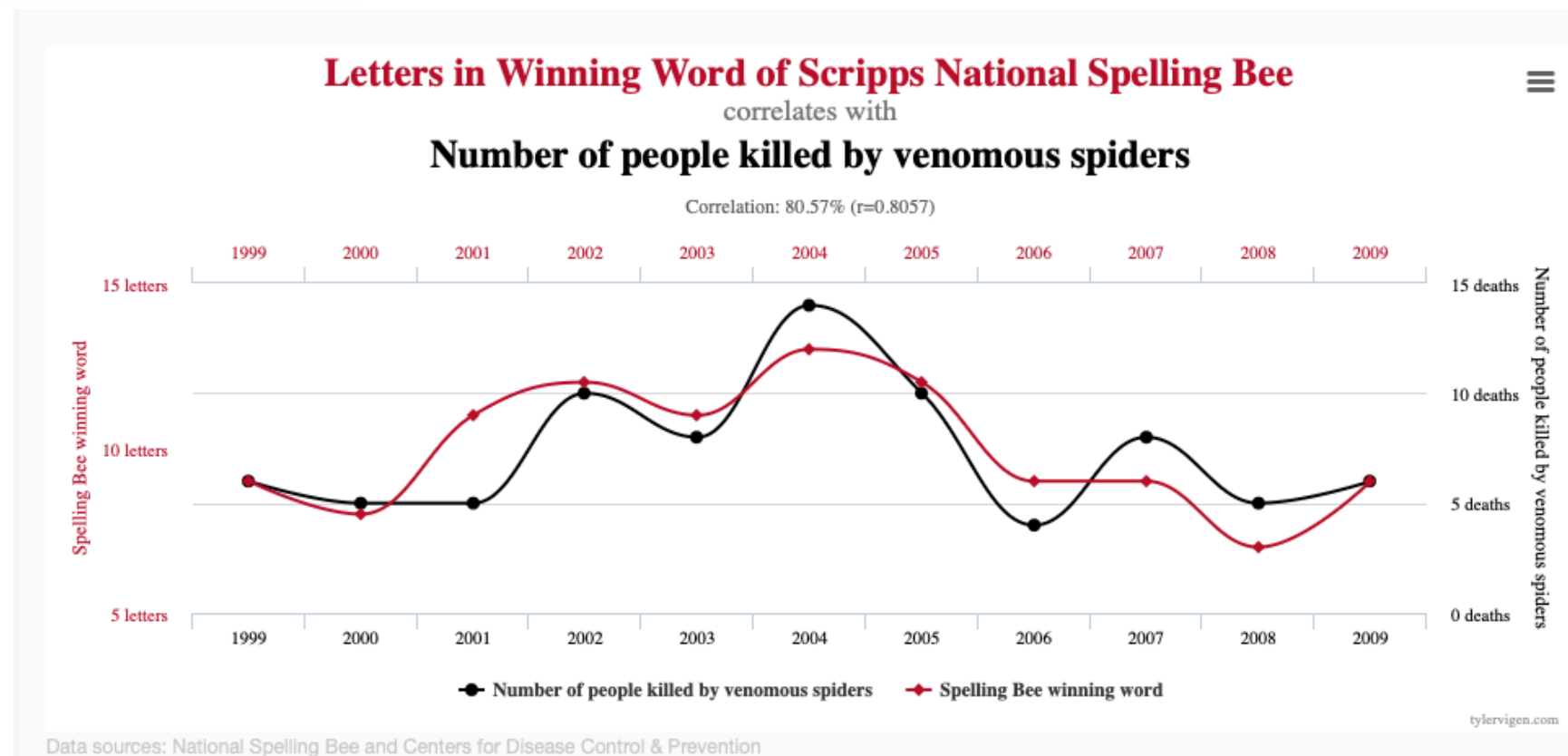
<https://xkcd.com/552/>

Do shark attacks cause more ice cream sales?

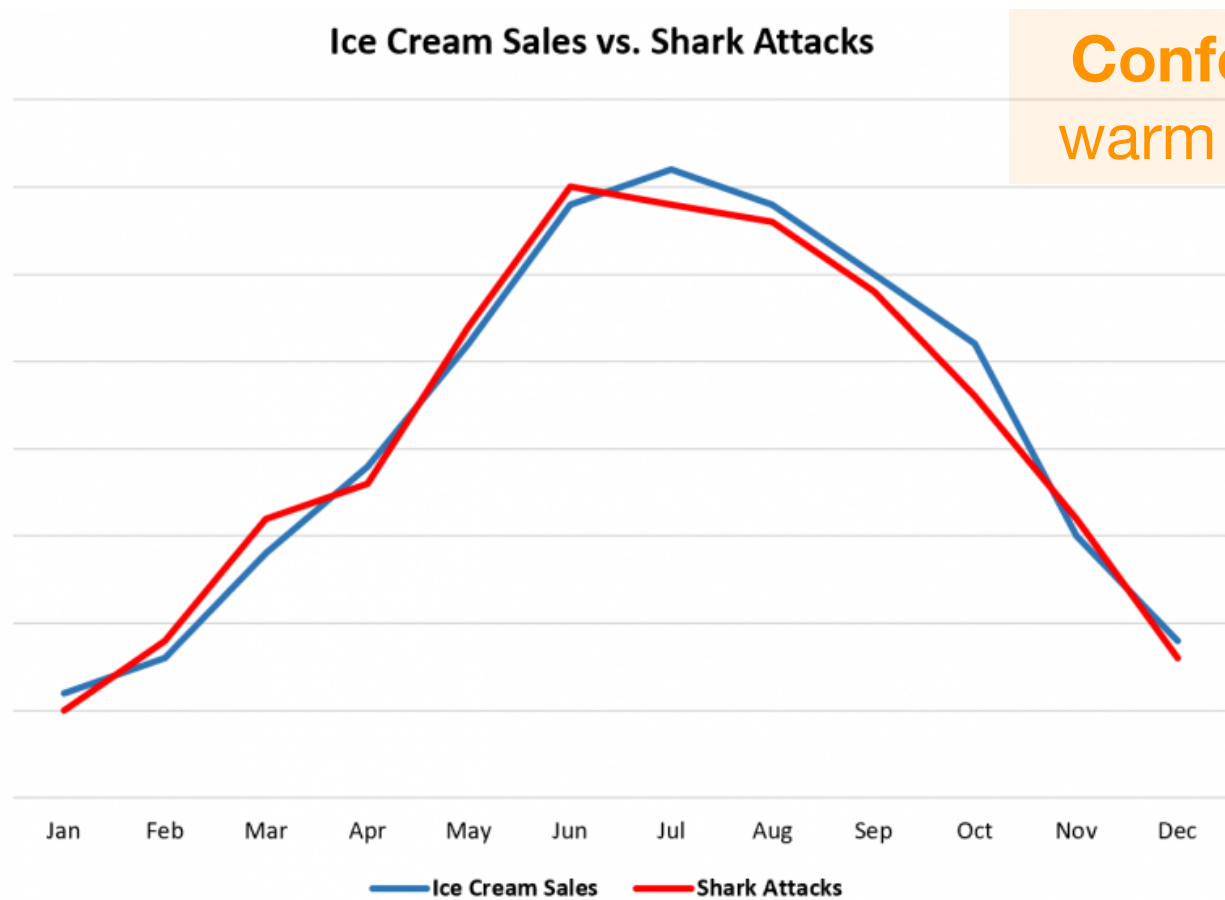


Do long spelling bee words cause venomous spider attacks?

<https://www.statology.org/correlation-does-not-imply-causation-examples/>



Do shark attacks cause more ice cream sales?

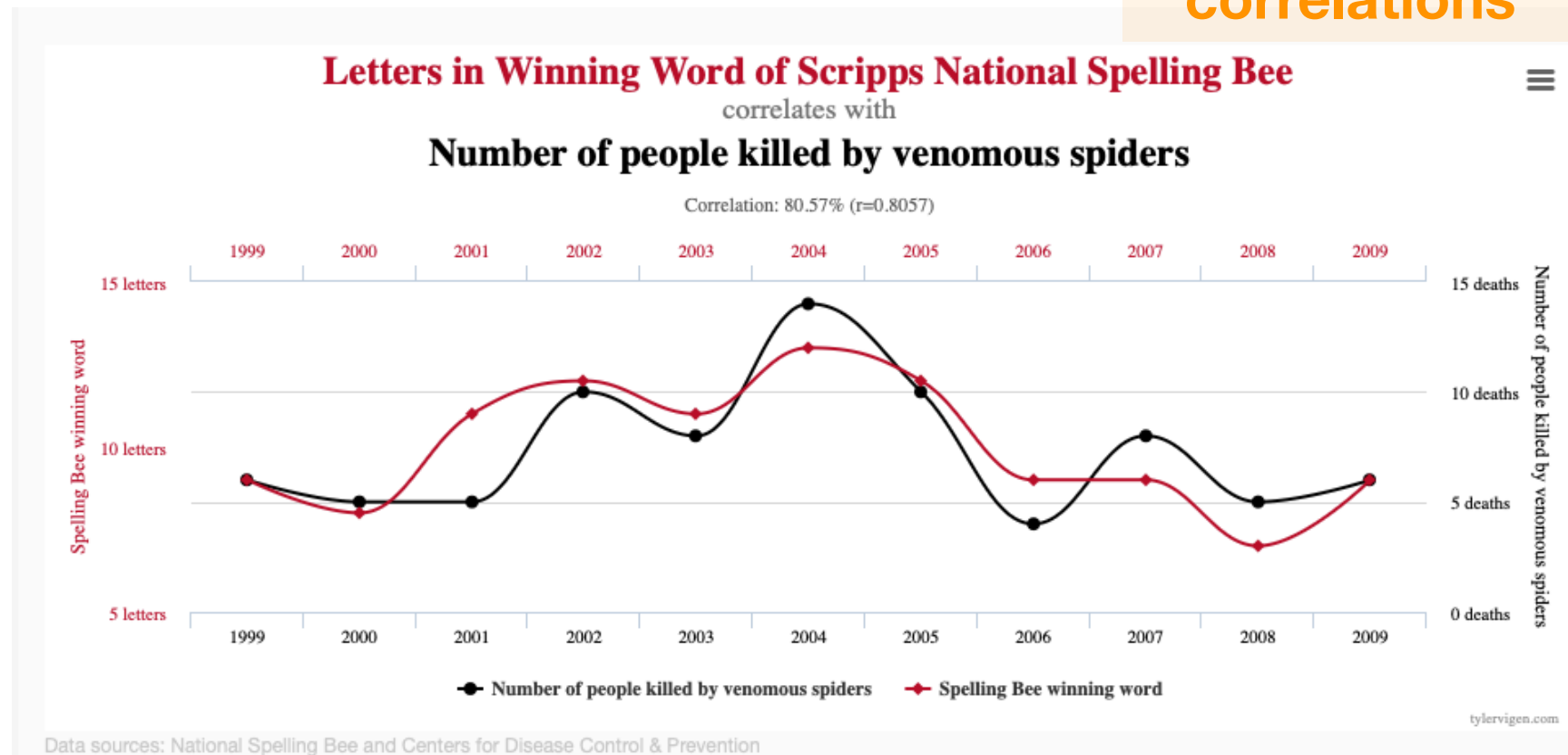


Confounder:
warm weather

Do long spelling bee words cause venomous spider attacks?

Cherry-picking correlations

<https://www.statology.org/correlation-does-not-imply-causation-examples/>



Remark: Prediction is not causation

(linear regression)

Consider linear regression.

Suppose we have the predictive model

$$Y_{\text{pred}} = \hat{\beta}_1 X_{\text{new},1} + \hat{\beta}_2 X_{\text{new},2} + \hat{\beta}_0.$$

We may want to know: does a change in $X_{\text{new},1}$ **cause** change in Y_{new} ?

In other words, if we change $X_{\text{new},1}$ while holding $X_{\text{new},2}$ fixed, how would Y_{new} change?

The coefficient $\hat{\beta}_1$ **DOES NOT** answer this question.

Suppose $\hat{\beta}_1 > 0$ and $\hat{\beta}_2 < 0$. The best interpretation we have is that, **in our training data**, samples where the first feature is large and second feature is small tend to have higher Y .

Remark: prediction is not causation

Concrete example (Card, 1993):

Each data point is a person.

Y = earning and X = (years in school).

Problem: Suppose “years in school” causes no change in “earning” but “parent-wealth” causes an increase in both “years in school” and “earning”.

Since we do not observe “parent-wealth”, we would see $\hat{\beta}_1 > 0$

Remark: Prediction is not causation

Consider linear regression.

Suppose we have the predictive model

$$Y_{\text{pred}} = \hat{\beta}_1 X_{\text{new},1} + \hat{\beta}_2 X_{\text{new},2} + \hat{\beta}_0.$$

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When can we make causal claims?

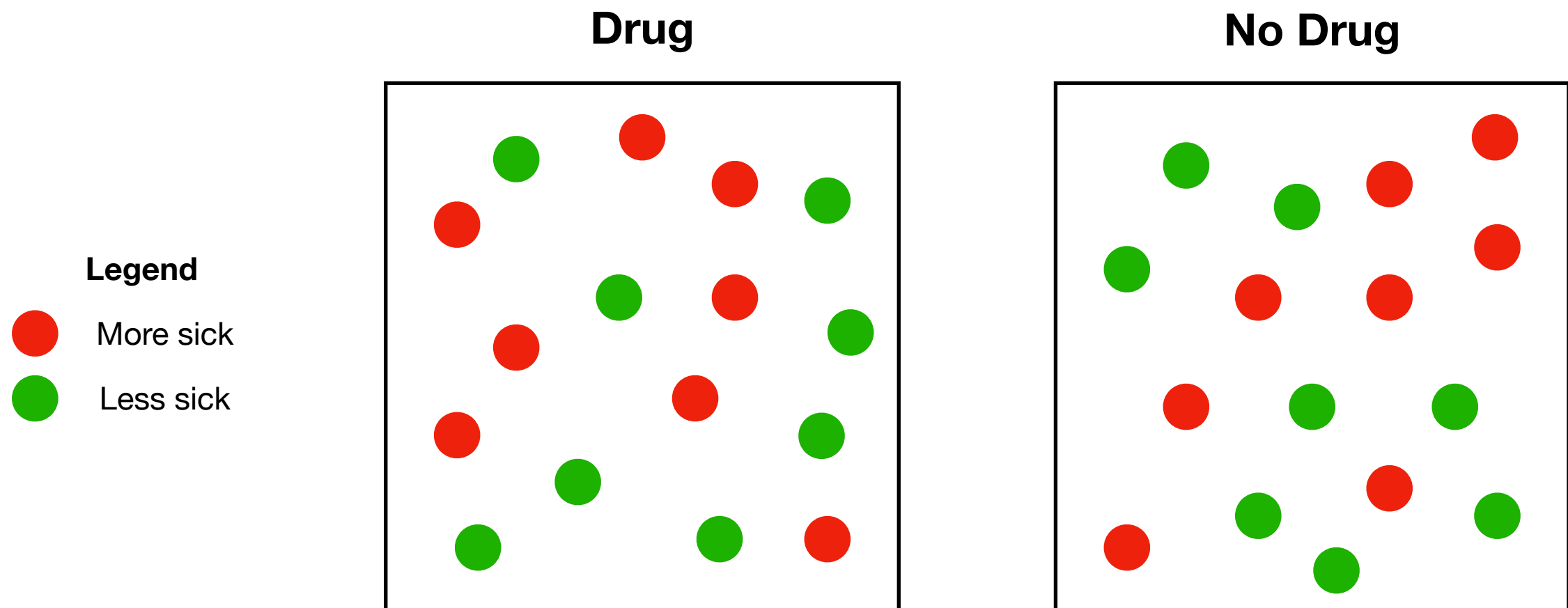
Outline

- Randomized Control Trials (RCTs) vs. Observational Studies
- Example: *Smoking and lung cancer*
- Intuition for causal inference methods in the absence of RCTs
- Potential Outcomes Framework
- Example: *Simpson's paradox*

We will see: carefully designed regression/classification, with some assumptions, is crucial for drawing causal conclusions!

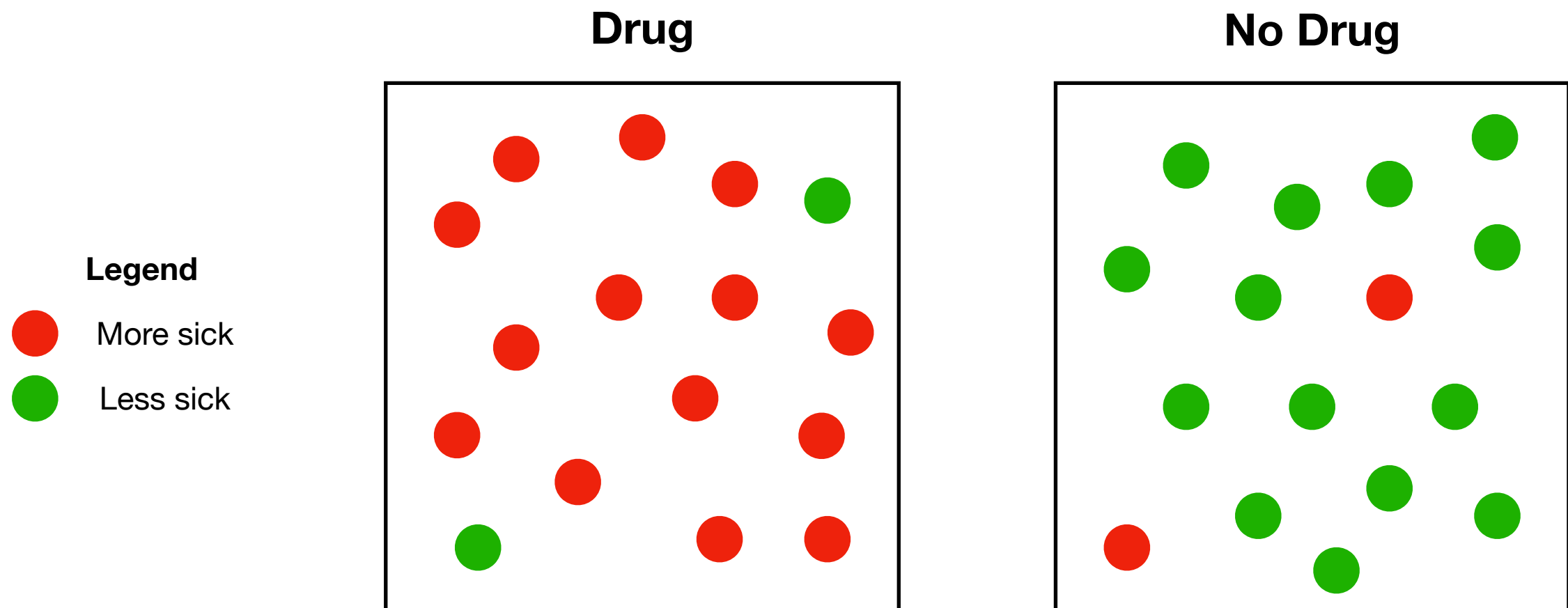
Example

- We have a new drug to treat a disease. *Does it increase lifespan?*
- **Gold standard:** Randomized Control Trial (RCT)
 - Randomly assign drug to patients, see if difference between drug/no drug groups is significant.



Example

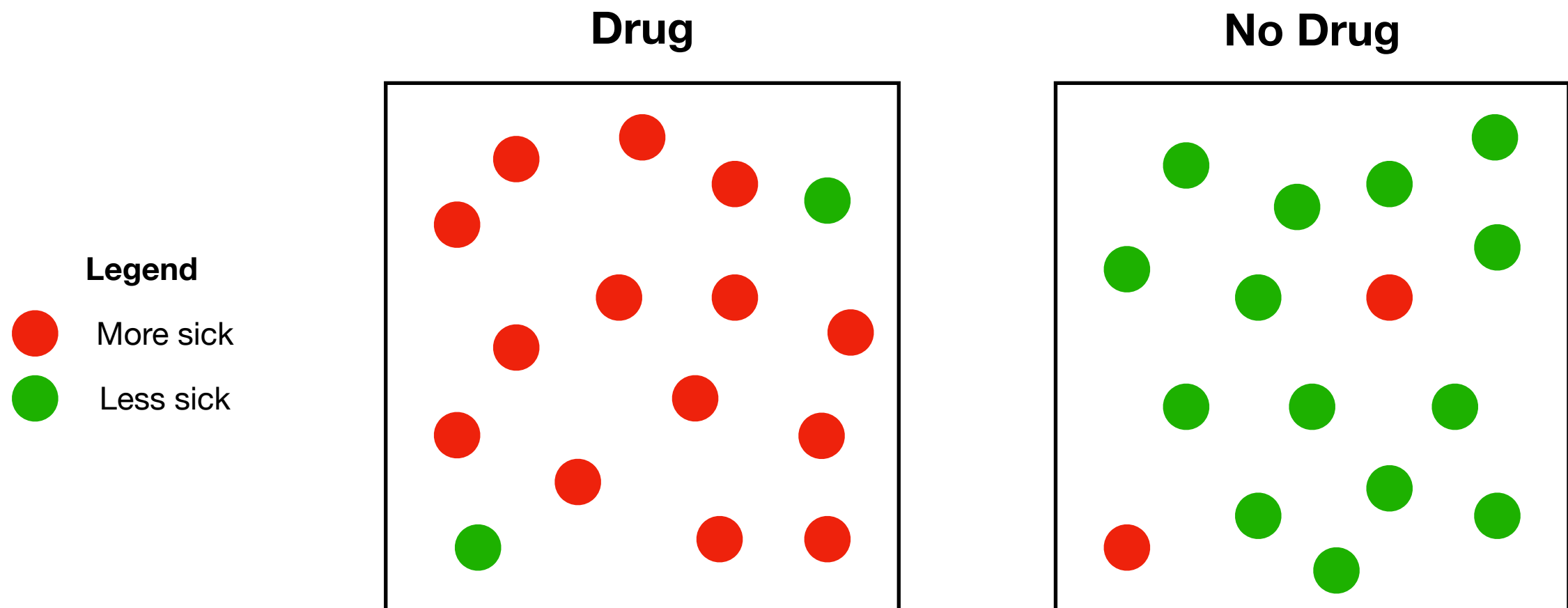
- We have a new drug to treat a disease. *Does it increase lifespan?*
- **Observational study:** no randomization of treatment...*what might go wrong?*
 - Doctors may give the drug more often to sicker patients...
 - when we measure the difference between groups, the effect of the drug on lifespan is *confounded* by the effect of patient sickness level.



Example

- We have a new drug to treat a disease. *Does it increase lifespan?*
- **Observational study:** no randomization of treatment. *What might go wrong?*
 - Doctors may give the drug more often to sicker patients.
 - when we measure the difference in lifespan, the effect of the drug on lifespan is *confounded* by the effect of patient sickness level.

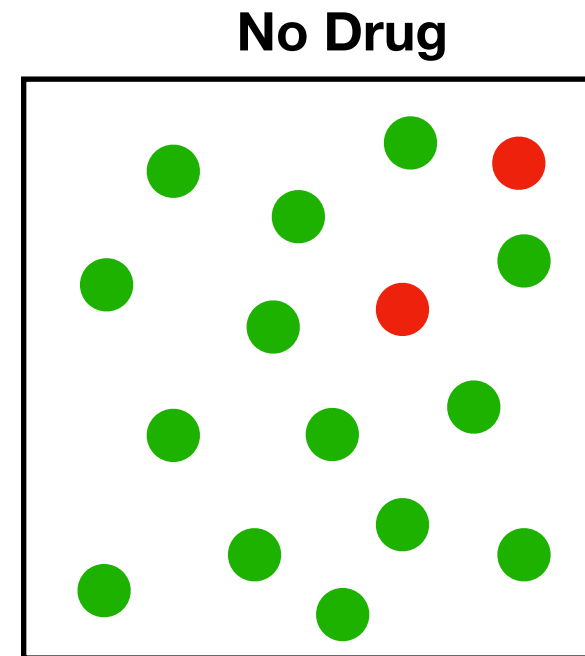
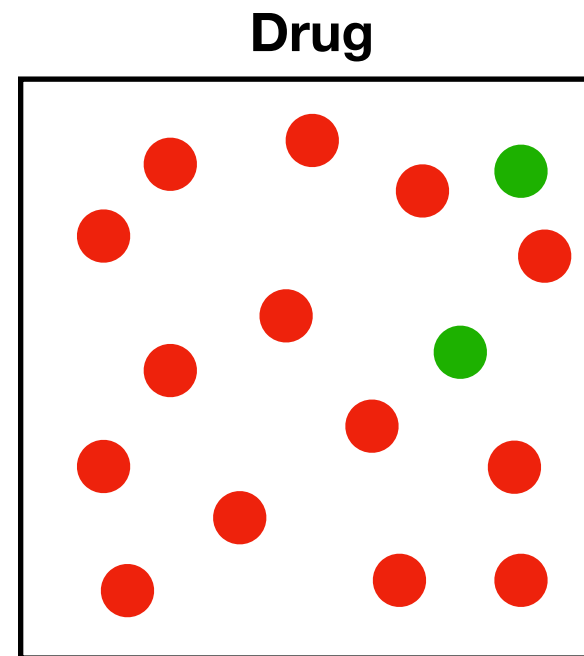
Is all hope lost?



Key idea

Observational Study

Legend
● More sick
● Less sick

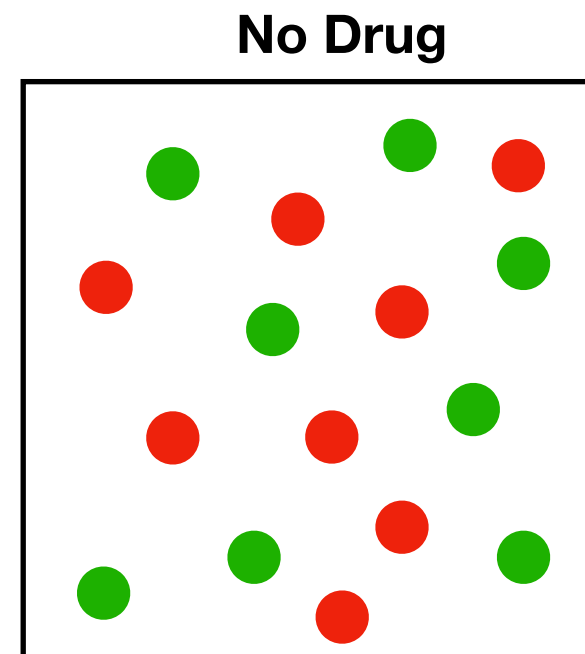
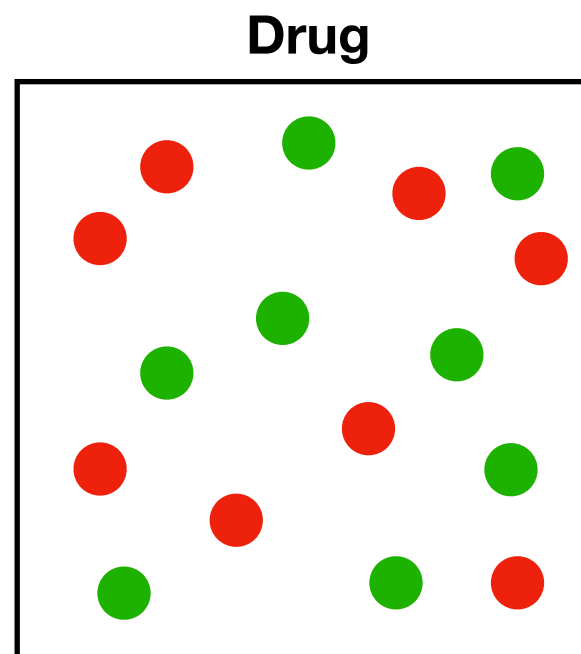


How can we “transform” our observational study into an RCT? **What assumptions do we need?**



Randomized Control Trial

Legend
● More sick
● Less sick



Example: Smoking

- Does smoking cause lung cancer?

1947

Every doctor in private practice was asked:
—family physicians, surgeons, specialists...
doctors in every branch of medicine—
“What cigarette do you smoke?”

According to a recent Nationwide survey:
More Doctors Smoke Camels
than any other cigarette!

Not a guess, not just a trend...but an actual fact based on the statements of doctors themselves to nationally known independent research organizations.

Yes, your doctor was asked...along with thousands of other doctors from Maine to California. And they've named their choice—the brand that more doctors named as their smoke is Camel! These nationally known independent research organizations found this to be a fact.

Nothing unusual about it. Doctors smoke for pleasure just like the rest of us. They appreciate, just as you, a mild taste that's cool and easy on the throat. They also enjoy a full, rich flavor of expertly blended golden tobacco. As they named Camels...more of them named Camels than any other brand. Next time you buy cigarettes, try Camels.

THE “T-ZONE” TEST WILL TELL YOU

The “T-Zone” is the taste and throat-feel zone of your mouth. It's the zone where your taste buds are most sensitive. It's the zone where you can tell the difference between good and bad cigarettes. It's the zone where you can tell the difference between a mild, cool Camel and a harsh, hot cigarette. So, the next time you buy cigarettes, try Camels. You'll see the difference.

“I’m going to grow a hundred years old!”

...and possibly she may—for the amazing strides of medical science have added years to life expectancy.

It's a fact—a warm and wonderful fact—that a 100-year-old child, or your own child, has a life expectancy almost a whole decade longer than was her mother's, and a good 10 to 20 years longer than that of her grandmother. Not only the expectancy of a longer life, but of a life by for healthiness. Thank medical science for that. Thank your doctor and thousands like him...making everything, often with little or no public recognition...that you and yours may enjoy a longer, better life.

According to a recent Nationwide survey:
More Doctors smoke Camels
than any other cigarette!

NOT ONE but three outstanding independent research organizations conducted this survey. And they asked not just a few thousand, but 181,507 doctors from coast to coast to name the cigarette they themselves preferred to smoke.

The answer came in by the thousands...from general physicians, diagnosticians, surgeons—yes, and men and women specialists too. The most-named brand was Camel.

If you are not now smoking Camels, try them. Compare them critically. See how the full, rich flavor of Camels' golden tobacco suits your taste. See how the cool richness of a Camel suits your throat. Let your “T-Zone” tell you just right.

CAMELS *Costlier Tobaccos*

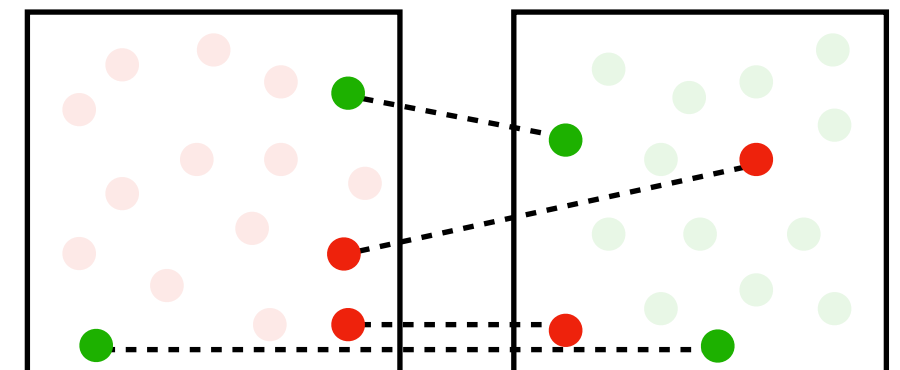
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- It would be unethical to run an RCT... we can't force people to smoke!
- How can we use observational data to answer this question?

Example: Smoking

- Hammond (*Journal of National Cancer Institute*) carried out observational studies, matching 36,975 heavy smokers to nonsmokers on the basis of:
 - age
 - race
 - nativity
 - rural versus urban residence
 - occupational exposures to dust and fumes
 - religion
 - education
 - marital status
 - alcohol consumption
 - sleep duration
 - exercise
 - severe nervous tension
 - use of tranquilizers
 - current health
 - family history of cancer other than skin cancer
 - family history of heart disease, stroke or high blood pressure
- Out of 36,975 pairs:
 - There were 122 pairs in which exactly one person died of lung cancer
 - **110** in which smoker died of lung cancer
 - **12** in which nonsmoker died of lung cancer



Solutions

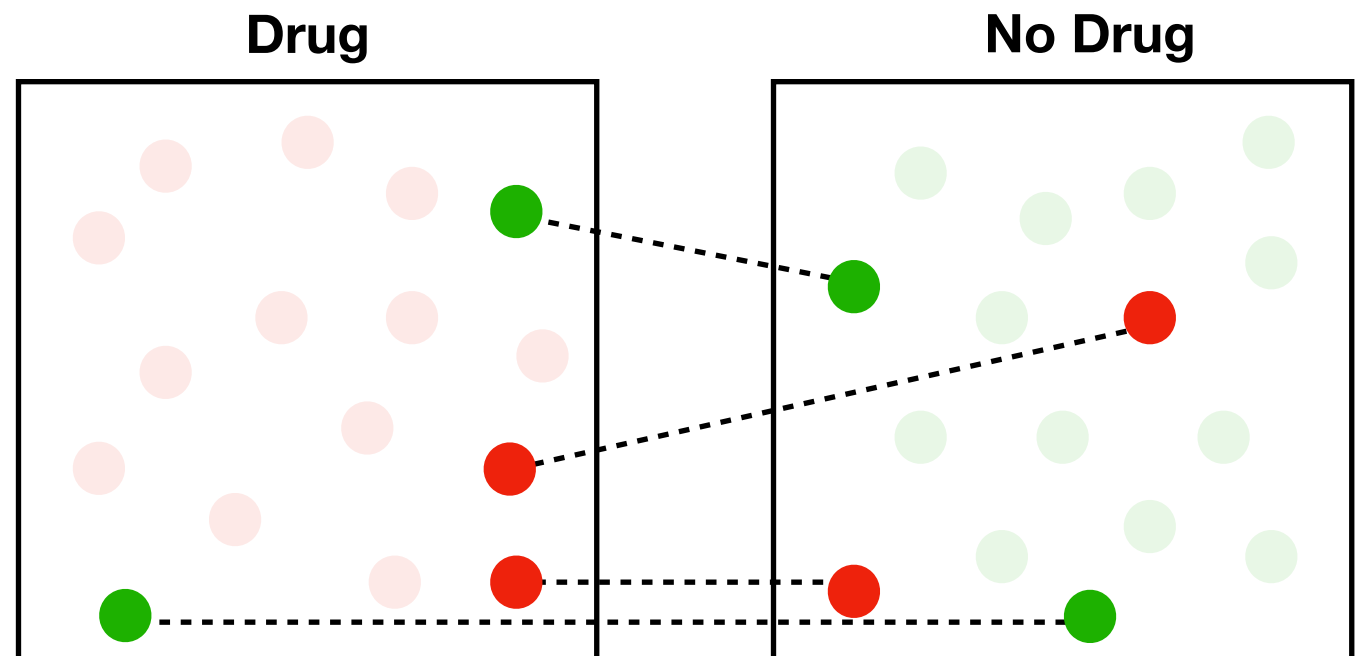
- Intuition for methods that can help measure *causal effects* in observational data

Matching

- Pair datapoints from each group that have same characteristics
- Calculate drug effect using difference between pairs

Legend

- More sick
- Less sick



Solutions

- Intuition for methods that can help measure *causal effects* in observational data

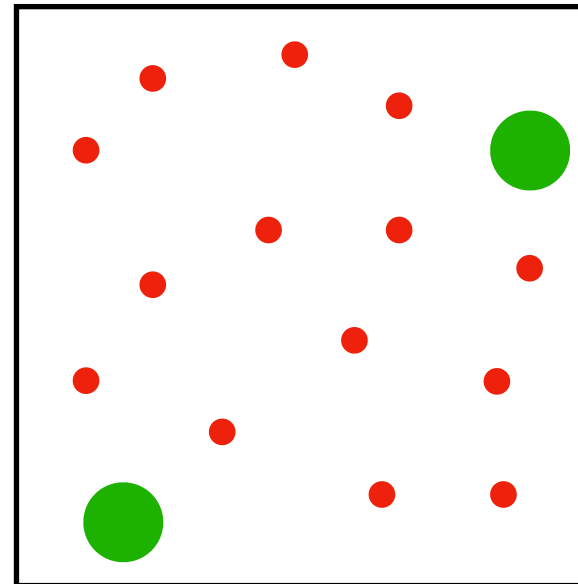
Propensity Score Reweighting

- Re-weight data so patient characteristics are more balanced when calculating the drug effect

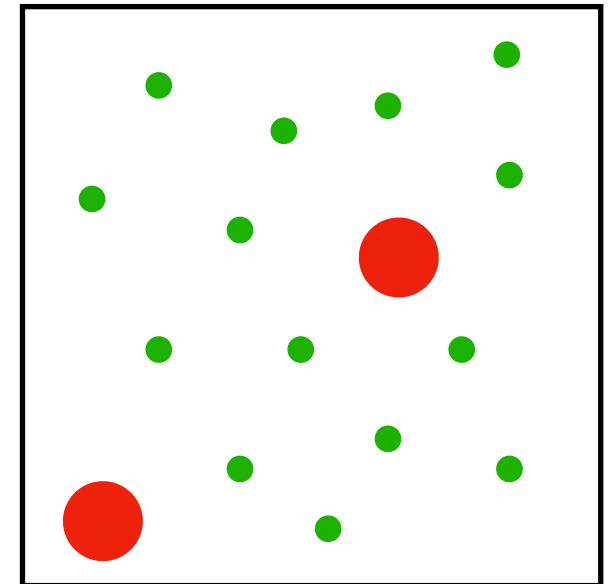
Legend

- More sick
- Less sick

Drug



No Drug



Solutions

- Intuition for methods that can help measure *causal effects* in observational data

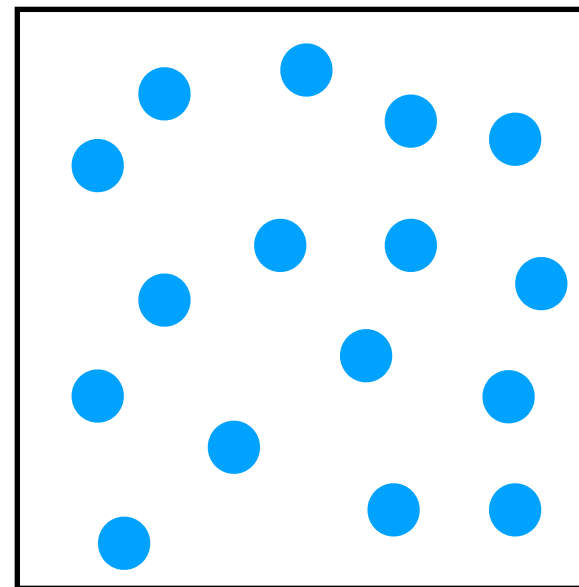
Adjustment (G-computation)

- Assume a model for how sickness affects lifespan, and adjust for sickness when calculating the drug effect

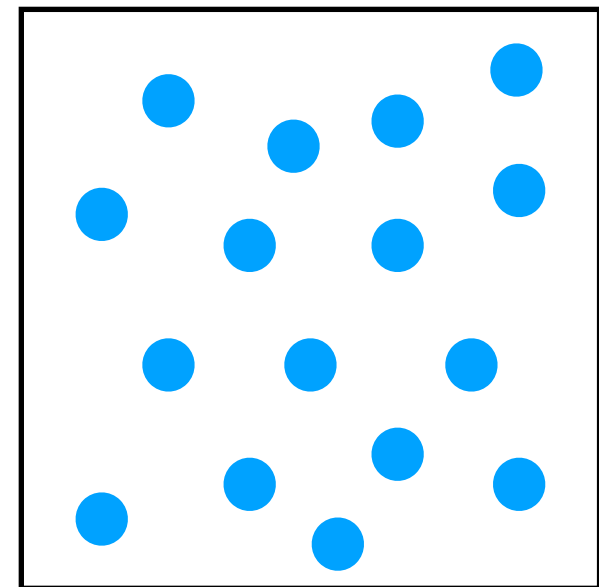
Legend

- More sick
- Less sick
- Adjusted

Drug



No Drug



The Potential Outcomes Framework

Data: we have n units and IID observations $\{(Y_i, T_i)\}_{i=1}^n$ where $Y_i \in \mathbb{R}$ is the response and $T_i \in \{0, 1\}$ is the treatment indicator.

$$T_i = 1 \quad \text{if unit } i \text{ received treatment.}$$

Potential outcome model: for each unit i , there exists

$$Y_i^{(0)} = \text{outcome for } i \text{ if } T_i = 0, \quad Y_i^{(1)} = \text{outcome for } i \text{ if } T_i = 1.$$

$$\text{We only observe } Y_i = \begin{cases} Y_i^{(0)} & \text{if } T_i = 0 \\ Y_i^{(1)} & \text{if } T_i = 1. \end{cases}$$

The Potential Outcomes Framework

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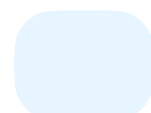
We only observe $Y_i = \begin{cases} Y_i^{(0)} & \text{if } T_i = 0 \\ Y_i^{(1)} & \text{if } T_i = 1. \end{cases}$

The fundamental problem of causal inference (Holland):

- It is impossible to observe both $Y_i^{(0)}$ and $Y_i^{(1)}$ and therefore it is impossible to observe the causal effect of the active treatment compared to the control for unit i

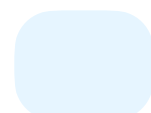
The Potential Outcomes Framework

Sample/ Individual	Outcome 1	Outcome 0	Treatment Assignment T
1	$Y_1^{(1)}$	$Y_1^{(0)}$	0
2	$Y_2^{(1)}$	$Y_2^{(0)}$	0
3	$Y_3^{(1)}$	$Y_3^{(0)}$	1
4	$Y_4^{(1)}$	$Y_4^{(0)}$	0
5	$Y_5^{(1)}$	$Y_5^{(0)}$	1
6	$Y_6^{(1)}$	$Y_6^{(0)}$	1
7	$Y_7^{(1)}$	$Y_7^{(0)}$	0
8	$Y_8^{(1)}$	$Y_8^{(1)}$	1
...			
N	$Y_n^{(1)}$	$Y_n^{(1)}$	0

 = observed

The Potential Outcomes Framework

Sample	Outcome 1	Outcome 0	Treatment Assignment T
1	$Y_1^{(1)}$	$Y_1^{(0)}$	0
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7	$Y_7^{(1)}$	$Y_7^{(0)}$	0
8	$Y_8^{(1)}$	$Y_8^{(1)}$	1
...			
N	$Y_n^{(1)}$	$Y_n^{(1)}$	0

 = observed

Individual treatment effects

$$Y_1^{(1)} - Y_1^{(0)}$$

$$Y_2^{(1)} - Y_2^{(0)}$$

$$Y_3^{(1)} - Y_3^{(0)}$$

...

$$Y_n^{(1)} - Y_n^{(0)}$$

Can't calculate!!!

Average Treatment Effects

Potential outcome model: for each unit i , there exists

$Y_i^{(0)}$ = outcome for i if $T_i = 0$, $Y_i^{(1)}$ = outcome for i if $T_i = 1$.

We only observe $Y_i = \begin{cases} Y_i^{(0)} & \text{if } T_i = 0 \\ Y_i^{(1)} & \text{if } T_i = 1. \end{cases}$

**What can we
measure
instead?**

Goal: estimate the average treatment effect (ATE)

$$\tau_0 = \mathbb{E}[Y_i^{(1)} - Y_i^{(0)}].$$

Average Treatment Effects

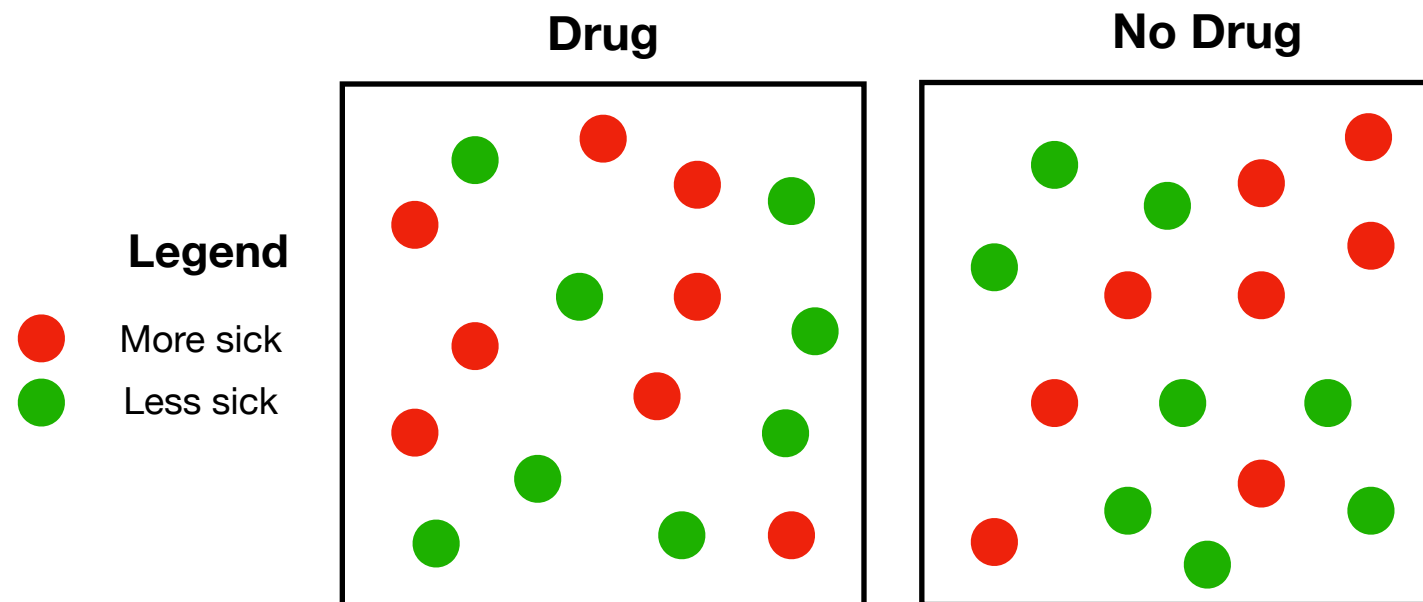
Goal: estimate the average treatment effect (ATE)

$$\tau_0 = \mathbb{E}[Y_i^{(1)} - Y_i^{(0)}].$$

When can we estimate ATE from data?

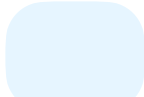
When treatment assignment T is independent of potential outcomes $Y_i^{(1)}$ and $Y_i^{(0)}$

In math: $Y_i^{(0)}, Y_i^{(1)} \perp T_i$



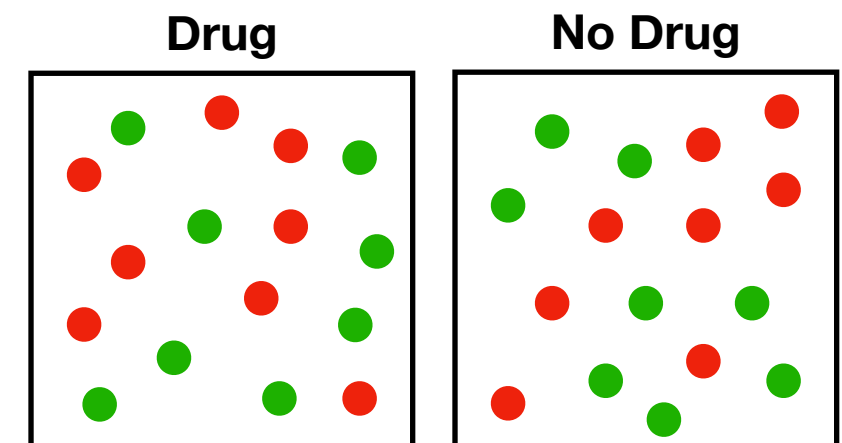
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8	$Y_8^{(1)}$	$Y_8^{(1)}$	1
...			
N	$Y_n^{(1)}$	$Y_n^{(1)}$	0

 = observed

$$Y_i^{(0)}, Y_i^{(1)} \perp T_i$$

In other words: knowing whether unit i was treated gives no information about what either of its potential outcomes would be



Legend

-  More sick
-  Less sick

Average Treatment Effects

Goal: estimate the average treatment effect (ATE)

$$\tau_0 = \mathbb{E}[Y_i^{(1)} - Y_i^{(0)}].$$

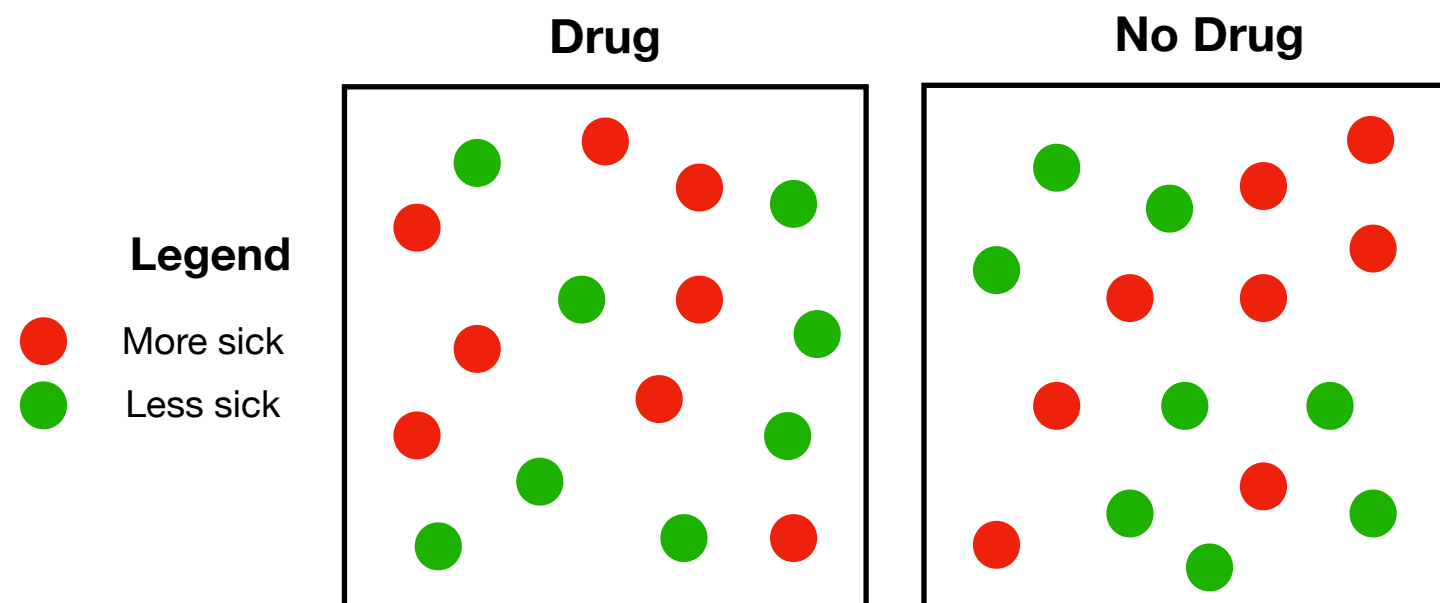
Randomized control trial (RCT) assumption:

$$T_i \sim \text{Bernoulli}(0.5)$$

$$Y_i^{(0)}, Y_i^{(1)} \perp T_i, \quad \text{so that}$$

$$\mathbb{E}[Y_i | T_i = 1] = \mathbb{E}[Y_i^{(1)} | T_i = 1] = \mathbb{E}[Y_i^{(1)}] \quad \text{and}$$

$$\mathbb{E}[Y_i | T_i = 0] = \mathbb{E}[Y_i^{(0)} | T_i = 0] = \mathbb{E}[Y_i^{(0)}].$$



Average Treatment Effects

Goal: estimate the average treatment effect (ATE)

$$\tau_0 = \mathbb{E}[Y_i^{(1)} - Y_i^{(0)}].$$

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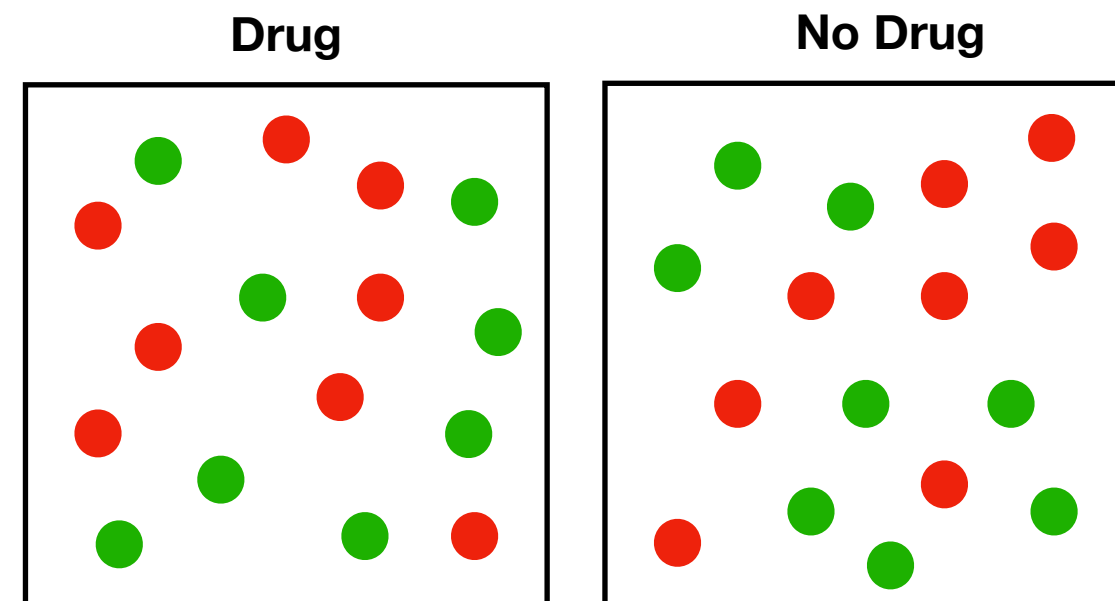
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$$\mathbb{E}[Y_i | T_i = 0] = \mathbb{E}[Y_i^{(0)} | T_i = 0] = \mathbb{E}[Y_i^{(0)}].$$

**What we can
calculate from
data**

**What we
want**

Legend
 More sick
 Less sick



Average Treatment Effects

Goal: estimate the average treatment effect (ATE)

$$\tau_0 = \mathbb{E}[Y_i^{(1)} - Y_i^{(0)}].$$

Estimator:

$$\hat{\tau} = \frac{1}{n_1} \sum_{i: T_i=1} Y_i - \frac{1}{n_0} \sum_{i: T_i=0} Y_i.$$

n_1 : number treated

n_0 : number not treated

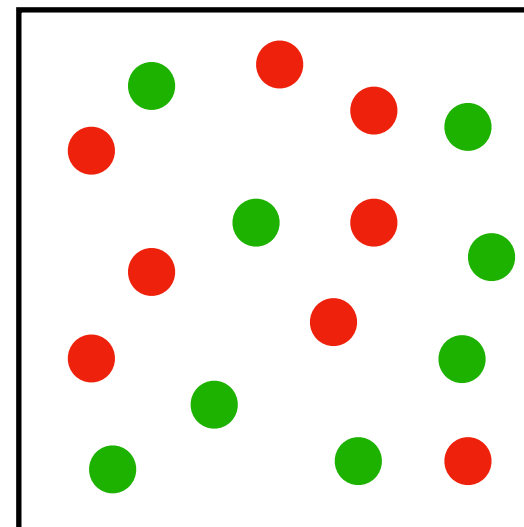
What is the effect of the drug?

Simple idea: take difference in means of Drug/NoDrug groups

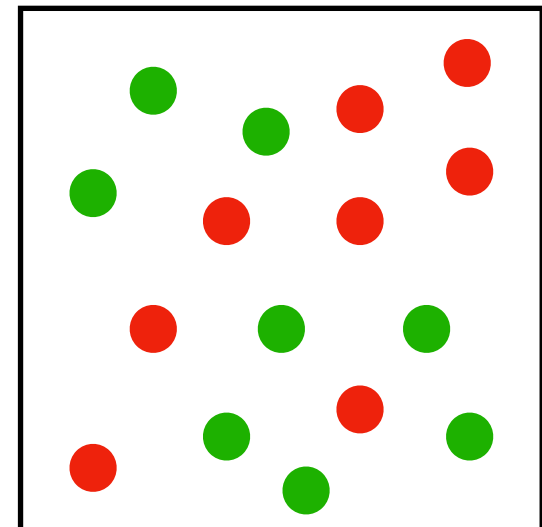
Legend

● More sick
● Less sick

Drug



No Drug



Let's see some code

Average Treatment Effects

Goal: estimate the average treatment effect (ATE)

$$\tau_0 = \mathbb{E}[Y_i^{(1)} - Y_i^{(0)}].$$

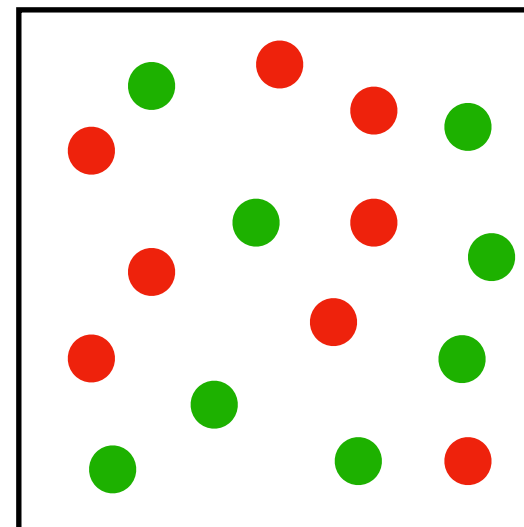
Remark 2. We note that $\hat{\tau}$ can also be obtained via OLS (although this is not obvious)

$$\hat{\tau} = \arg \min_{\tau \in \mathbb{R}} \min_{\mu \in \mathbb{R}} \sum_{i=1}^n (Y_i - \tau T_i - \mu)^2.$$

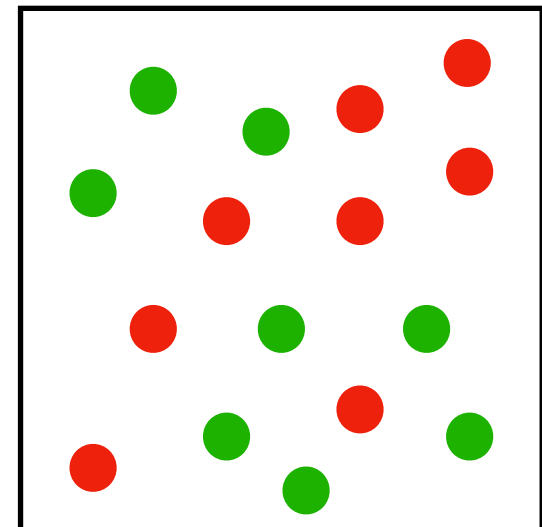
Legend

● More sick
● Less sick

Drug



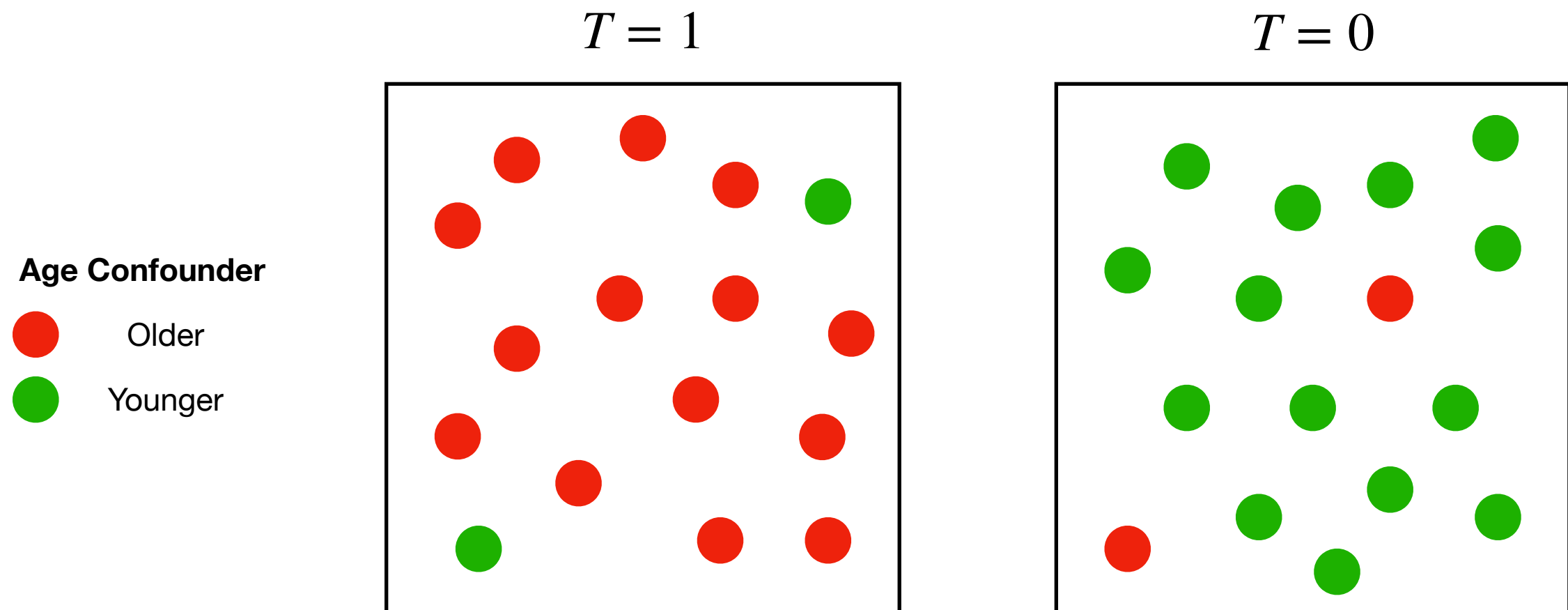
No Drug



The Potential Outcomes Framework

- In the potential outcomes framework, what happens if we **do not** have an RCT? i.e we are in **an observational** setting

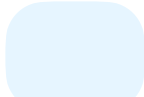
Example 1. Without the RCT assumption, we may have $\mathbb{E}\hat{\tau} \neq \tau$.



If we take the difference of means, we may just get the effect of age, not the treatment T !

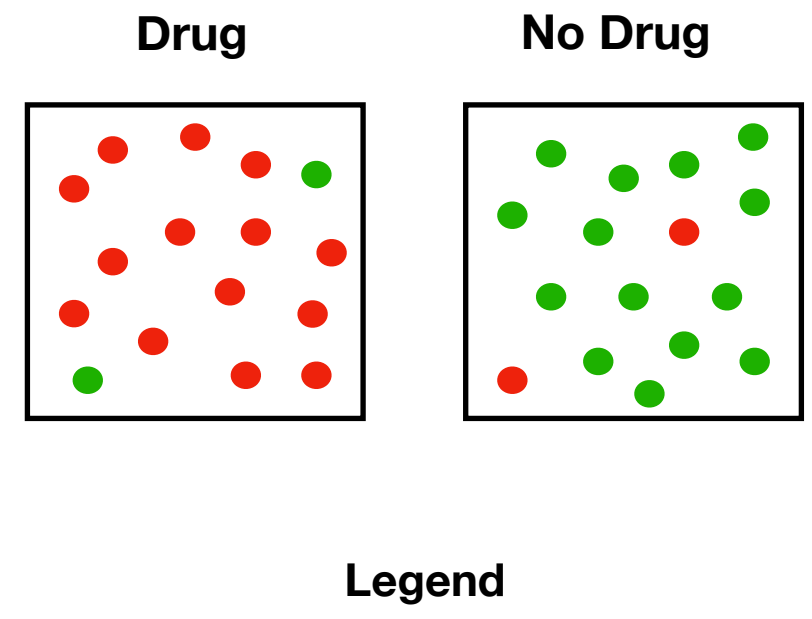
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Sample	Outcome 1	Outcome 0	Treatment Assignment T
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7	$Y_7^{(1)}$	$Y_7^{(0)}$	0
8	$Y_8^{(1)}$	$Y_8^{(1)}$	1
...			
N	$Y_n^{(1)}$	$Y_n^{(1)}$	0

 = observed

$$Y_i^{(0)}, Y_i^{(1)} \not\leq T_i$$

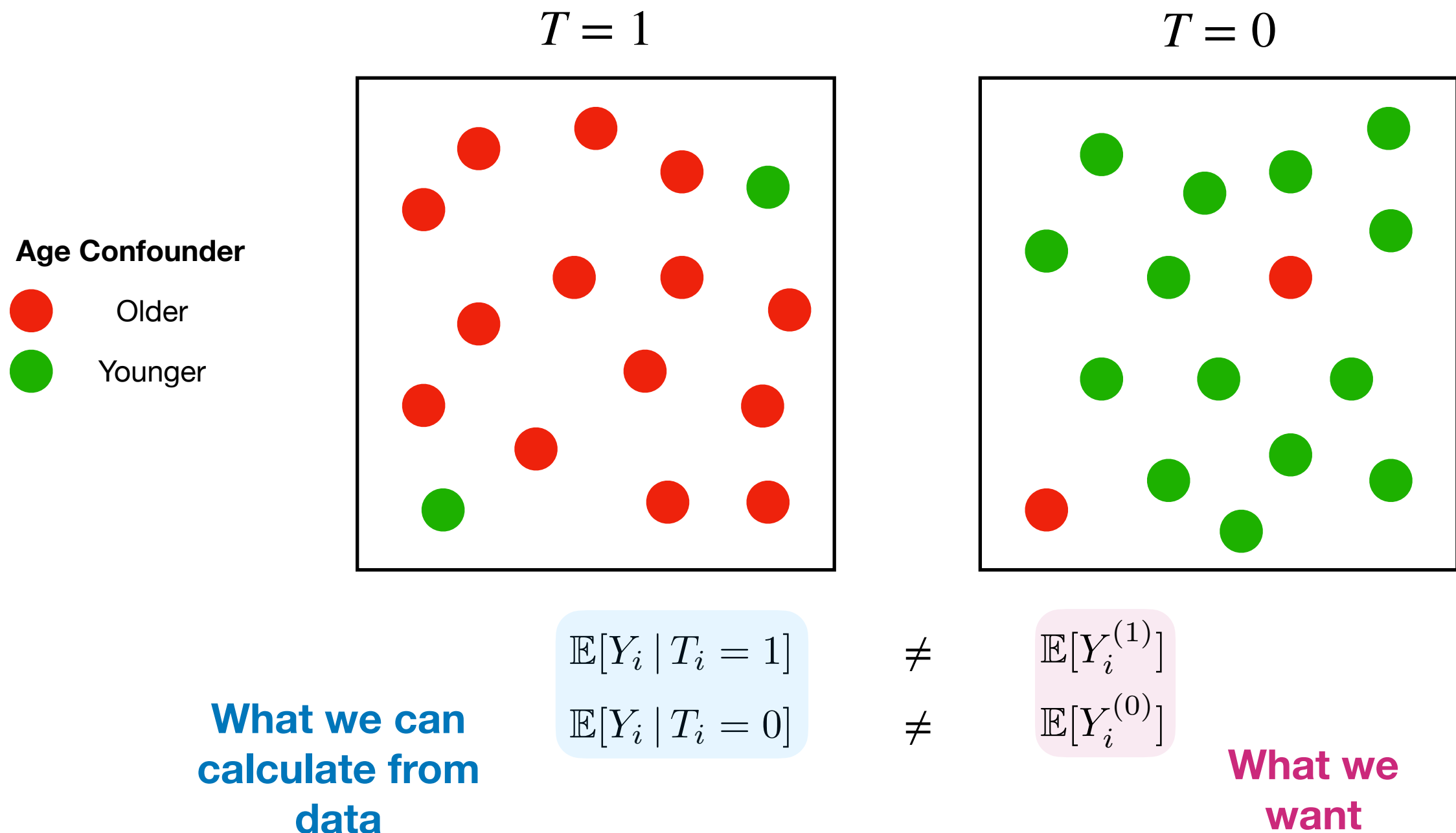
Example: we only see $Y_i^{(1)}$ if the actual $Y_i^{(1)}$ value is smaller (more sick patients)



What if we are in an observational setting?

- We **do not** have an RCT

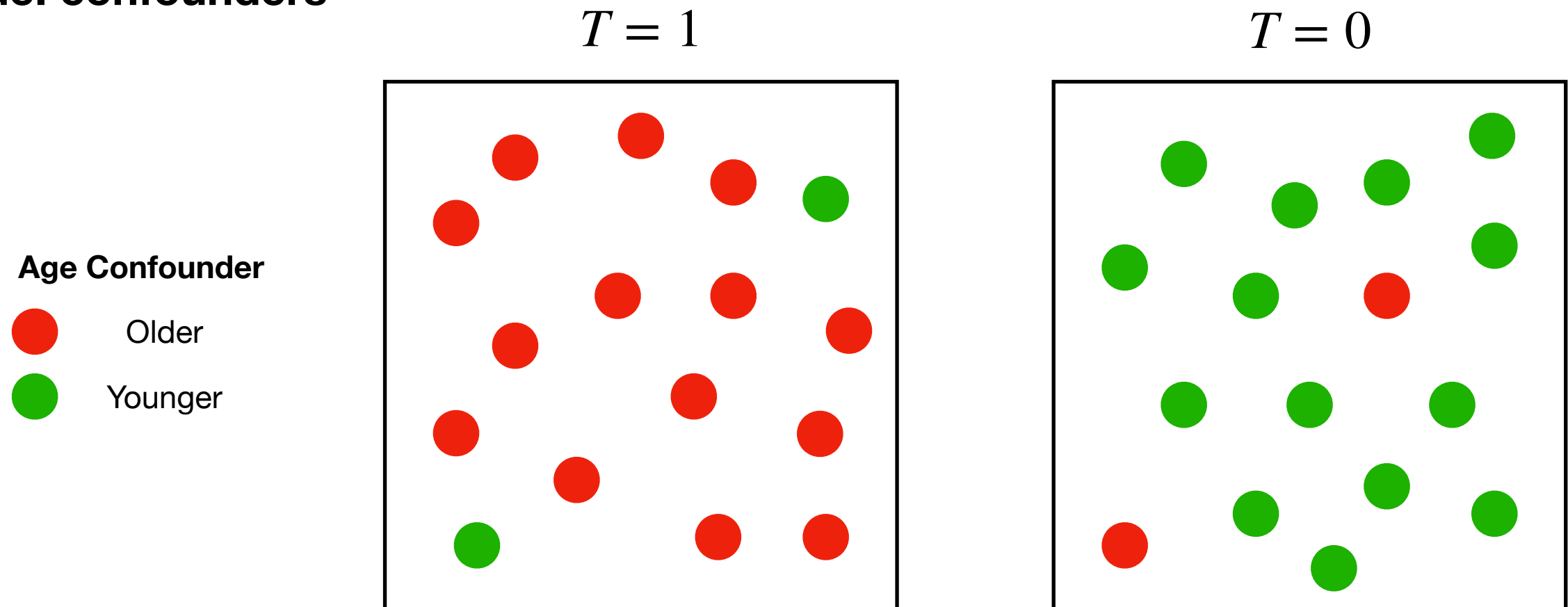
Example 1. Without the RCT assumption, we may have $\mathbb{E}\hat{\tau} \neq \tau$.



Why is RCT failing and what can we do

$$Y_i^{(0)}, Y_i^{(1)} \not\perp T_i$$

Issue: confounders



Solution: include features (confounders) in the analysis

- We should include some features, not necessarily all, as described in next two paradoxes.

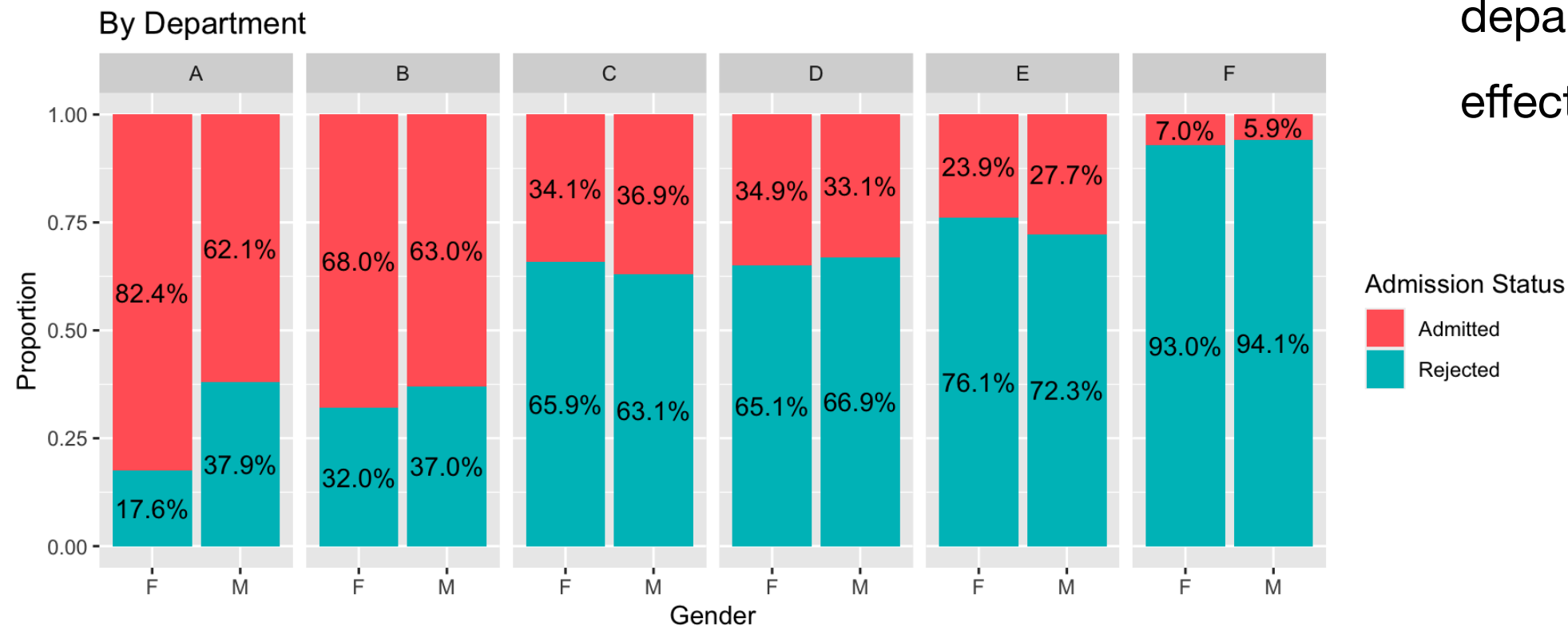
Simpson's Paradox

- Suppose we want to understand the causal effect of being a woman on college admission

- It looks like women are admitted at a much lower rate than men (30.4% vs. 44.5%)

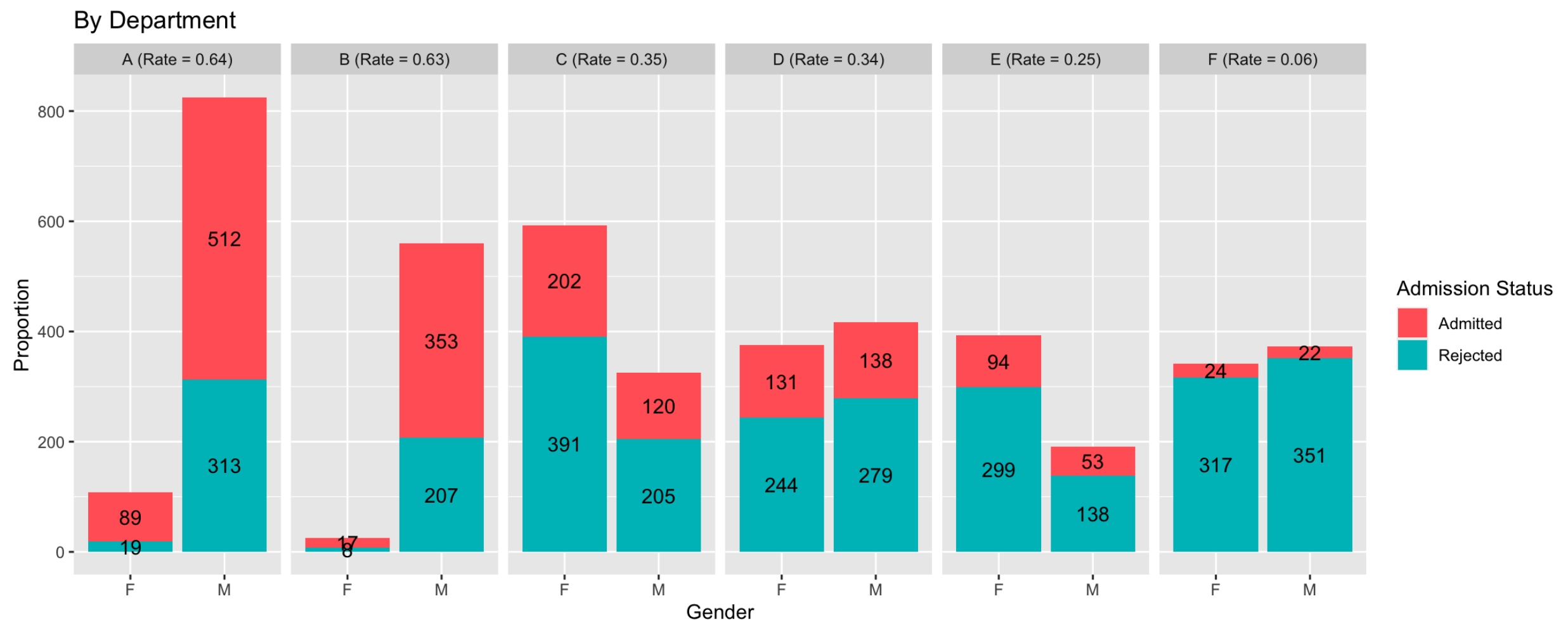


- If you separate by department, however, this effect disappears



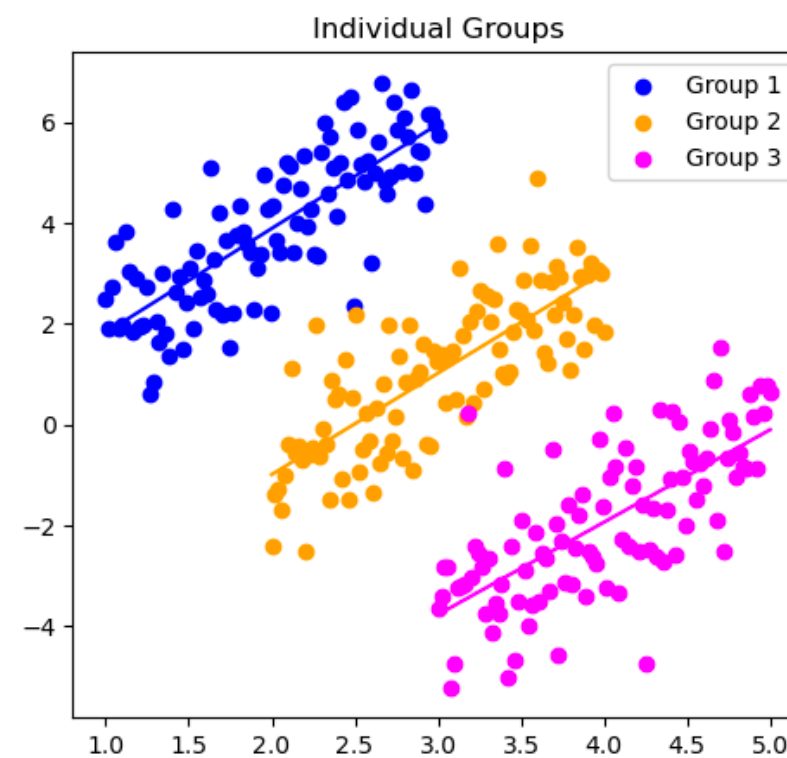
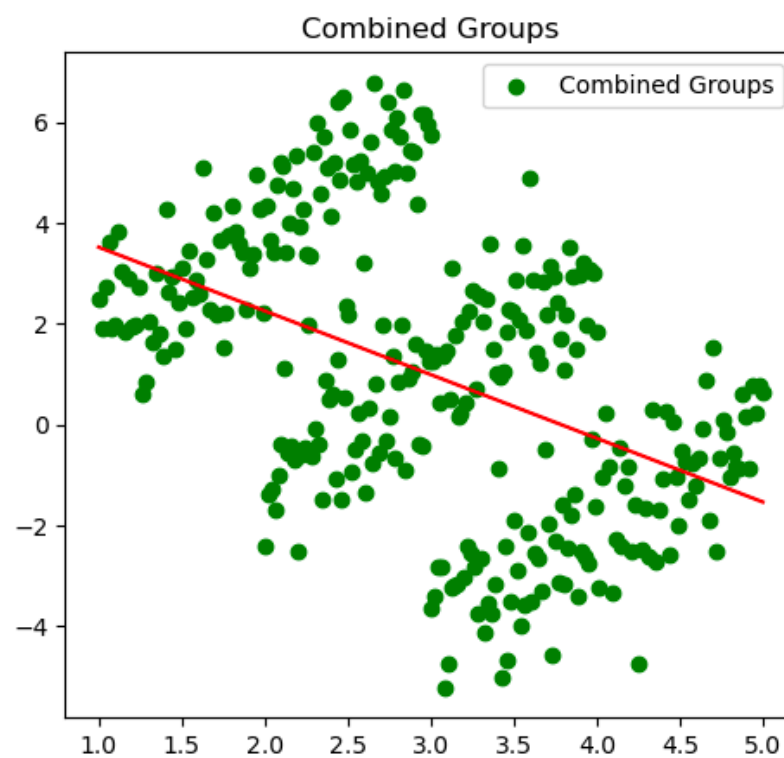
Simpson's Paradox

- Suppose we want to understand the causal effect of being a woman on college admission
- What's happening?
 - Many fewer women applied to departments with high acceptance rates



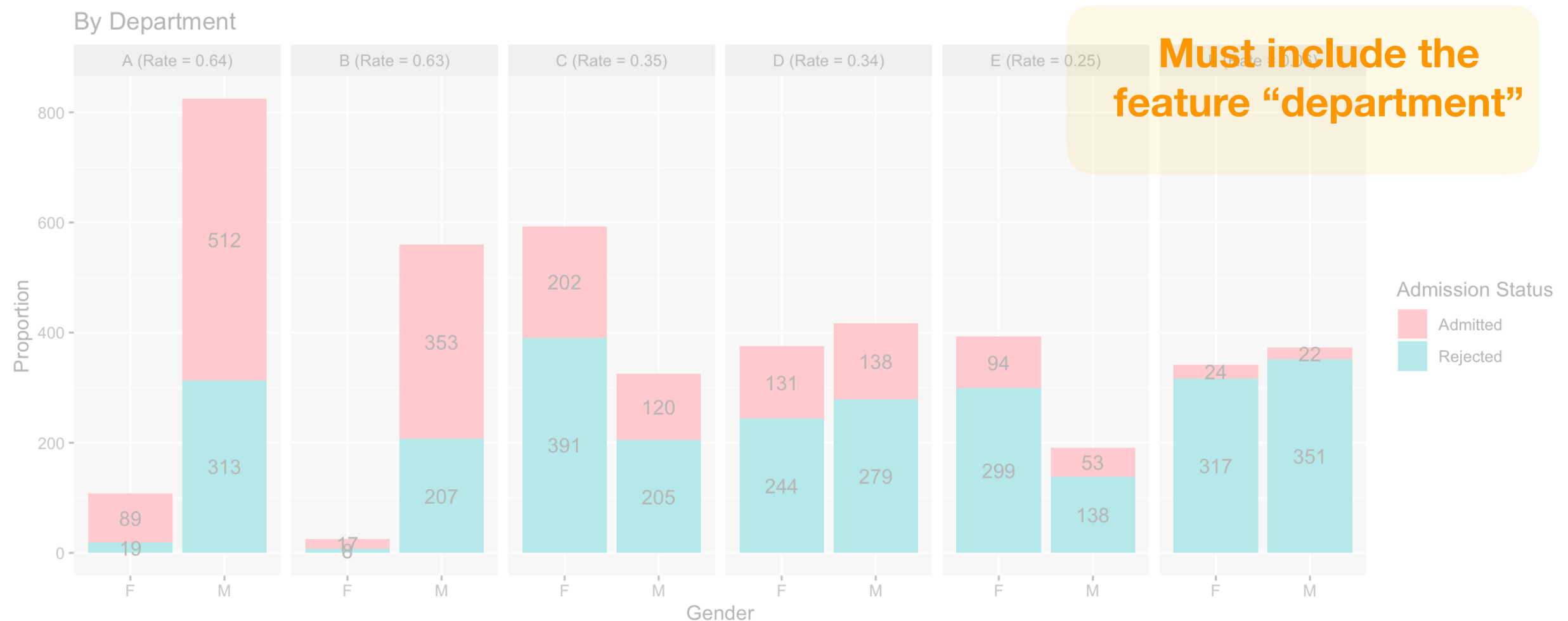
Simpson's Paradox

- Effect can reversing, after conditioning on a third variable “department”



Simpson's Paradox

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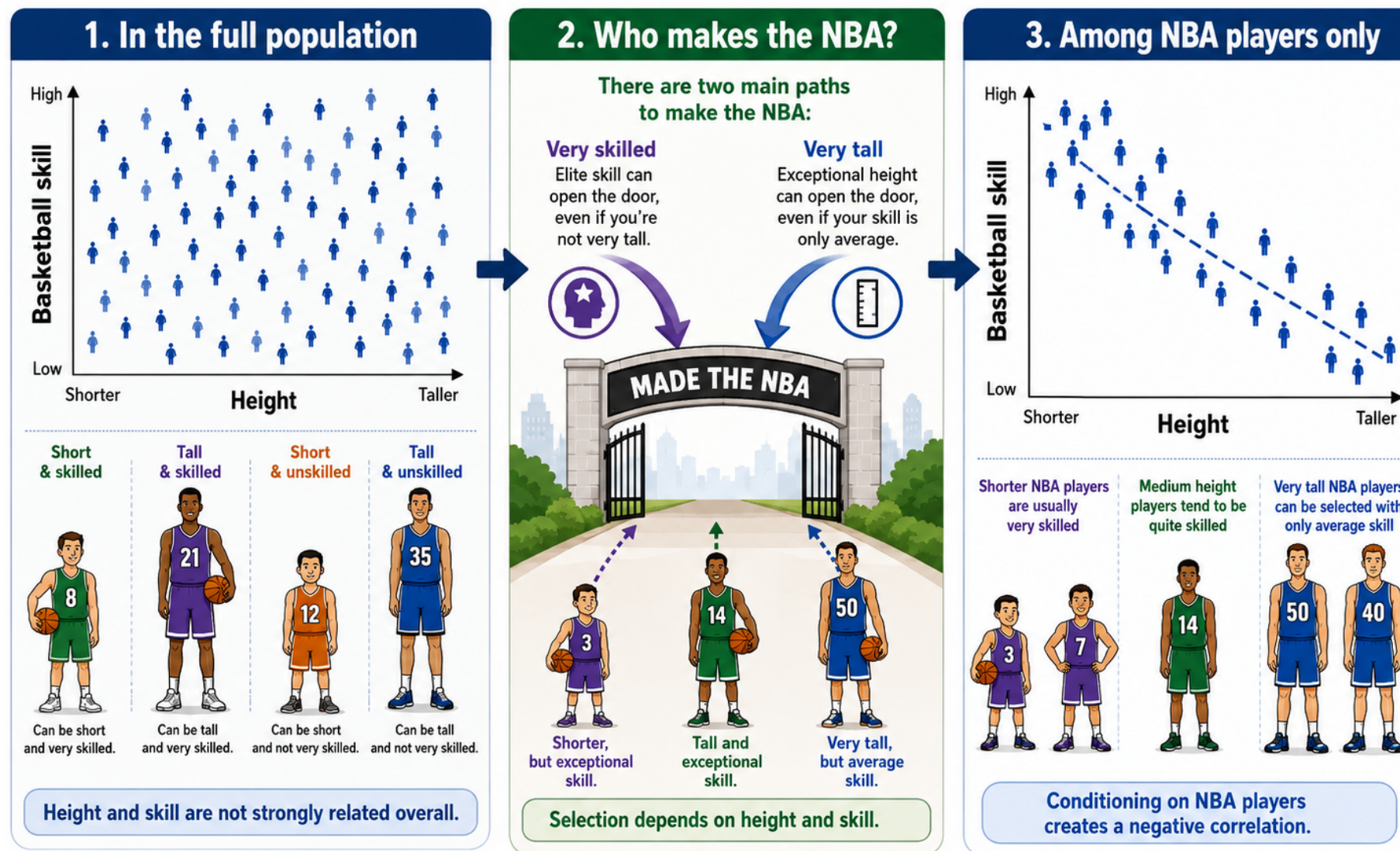


Berkson's Paradox (aka collider-bias)

- Suppose we want to understand the causal effect of height on skill at Basketball
- We observe a negative correlation between height and skill among NBA players
- Conclusion we **wrongly make**: Being taller means less skillful at Basketball

Berkson's Paradox (aka collider-bias)

- Suppose we want to understand the causal effect of height on skill at Basketball
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Berkson's paradox

If you only look at people who made the NBA, height and skill can look negatively related — even if they were not negatively related in the full population. Shorter players usually need more skill to be selected, while very tall players can be selected with less skill.



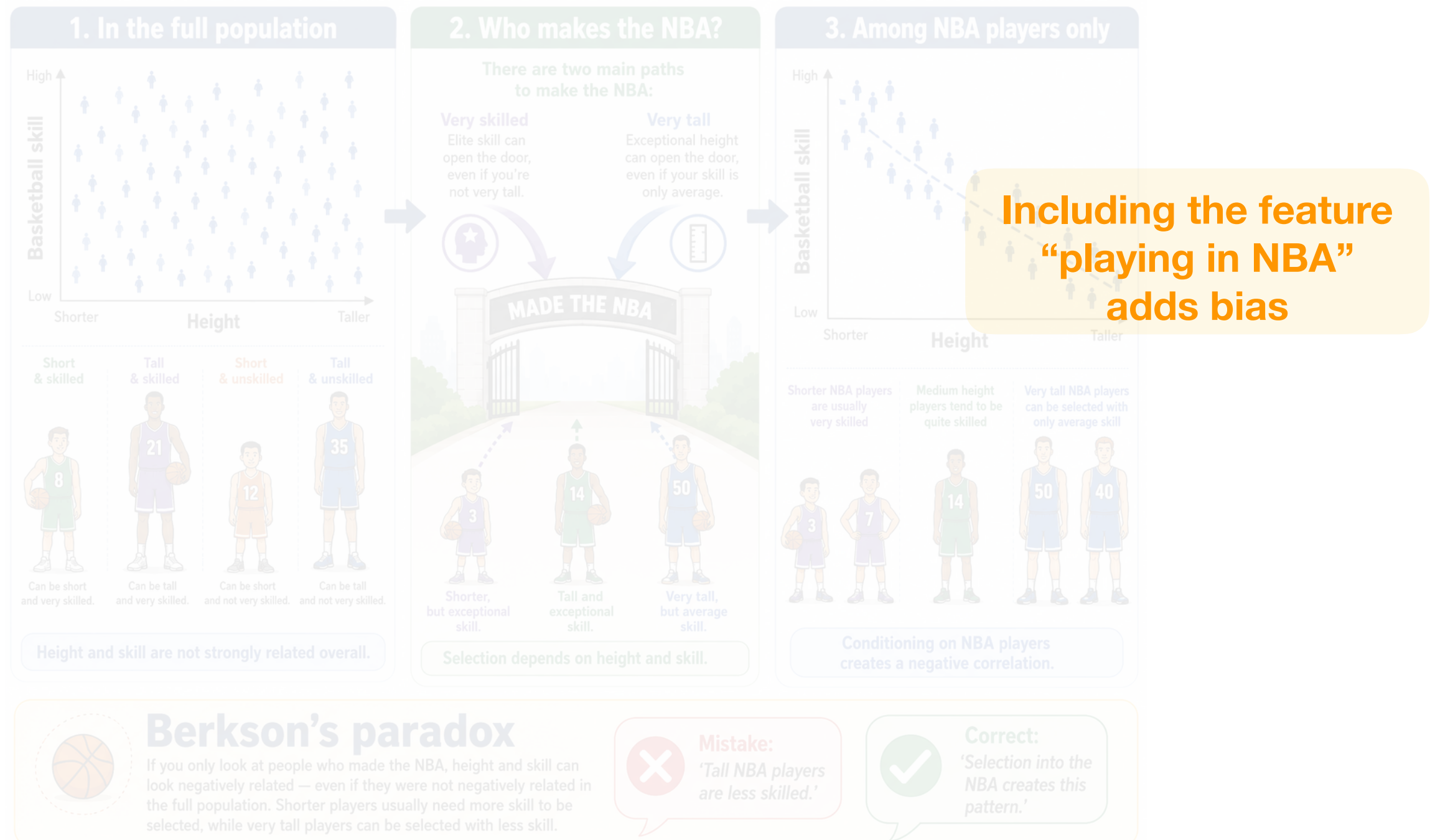
Mistake:
'Tall NBA players are less skilled.'



Correct:
'Selection into the NBA creates this pattern.'

Berkson's Paradox (aka collider-bias)

- Suppose we want to understand the causal effect of height on skill at Basketball
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Observed Confounders

Data: suppose we observe $\{(Y_i, T_i, X_i)\}_{i=1}^n$ where $Y_i \in \mathbb{R}$ is the response, $T_i \in \{0, 1\}$ is the treatment indicator, and $X_i \in \mathbb{R}^p$ is the observed features.

RCT assumption

$$Y_i^{(0)}, Y_i^{(1)} \perp T_i$$

Observational assumption:

$$Y_i^{(0)}, Y_i^{(1)} \perp T_i \mid X_i.$$

In other words: among units with the same covariates, knowing whether the unit was treated gives no information about what either of its potential outcomes would be

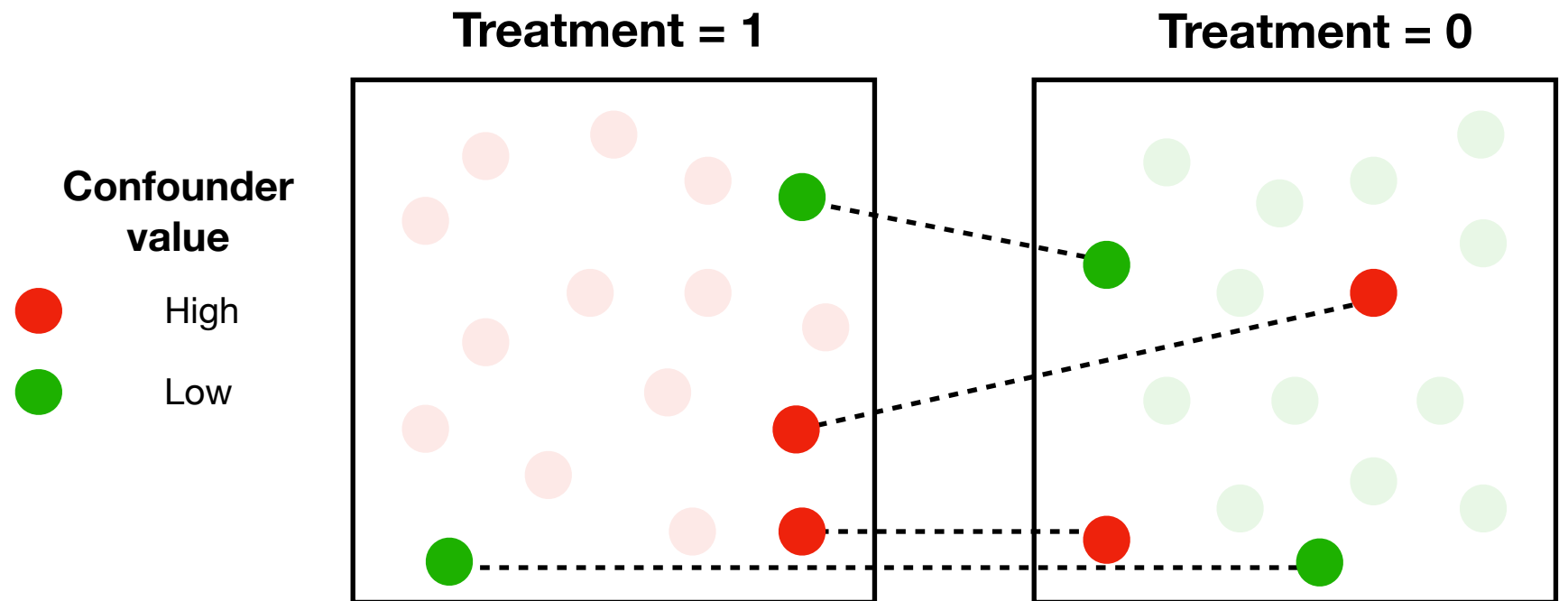
Note: observational assumption may hold over some of the covariates (covariate selection is important!)

Observed Confounders

$$Y_i^{(0)}, Y_i^{(1)} \perp T_i \mid X_i.$$

Matching

- Pair datapoints from each group that have similar X
- Calculate treatment effect using difference between pairs



Observed Confounders

Propensity Score Reweighting

- Reweight data so unit characteristics are more balanced when calculating the treatment effect

Propensity score

Definition: for $x \in \mathbb{R}^p$, define

$$q(x) := \mathbb{P}(T_i = 1 \mid X_i = x).$$

We refer to $q(\cdot)$ as the **propensity score**.

Inverse probability weighted estimator: suppose we have an estimator $\hat{q}(\cdot)$ of the propensity score, we may define

$$\hat{\tau}_{\text{ipw}} = \frac{1}{n} \sum_{i=1}^n \frac{T_i Y_i}{\hat{q}(X_i)} - \frac{(1 - T_i) Y_i}{1 - \hat{q}(X_i)}.$$

Observed Confounders

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Note: $q(X)$ here is the truth, which is not known in practice

Estimating propensity score

Average treatment effect

$$\hat{\tau}_{\text{ipw}} = \frac{1}{n} \sum_{i=1}^n \frac{T_i Y_i}{\hat{q}(X_i)} - \frac{(1 - T_i) Y_i}{1 - \hat{q}(X_i)}$$

$$\hat{q}(X_i) = \widehat{\mathbb{P}}(T_i = 1 | X_i)$$

estimated probability of receiving
a treatment given particular features

Q. How can we use the tools from class to learn these probabilities?

Estimating propensity score

Average treatment effect

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estimated probability of receiving
a treatment given particular features

Q. How can we use the tools from class to learn these probabilities?

A. Logistic regression model! We could also use more complex classification models like trees

Let's see some code

G-computation: an adjustment approach

Goal: estimate $\tau_0 = \mathbb{E}[Y_i^{(1)} - Y_i^{(0)}]$

Regress Y on T and X:

$$\hat{Y} = \hat{m}(X, T)$$

A regression model:

- linear regression
- trees
- Neural networks

Example: $\hat{Y} = \hat{\beta}_1 X + \hat{\beta}_2 T$

G-computation: an adjustment approach

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A regression model:

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Obtain potential outcome estimates

$$\hat{Y}_i^{(0)} = \hat{m}(X_i, T_i = 0)$$

$$\hat{Y}_i^{(1)} = \hat{m}(X_i, T_i = 1)$$

Example: $\hat{Y}_i^{(0)} = \hat{\beta}_1 X_i + \hat{\beta}_2 \cdot 0$

Example: $\hat{Y}_i^{(1)} = \hat{\beta}_1 X + \hat{\beta}_2 \cdot 1$

G-computation: an adjustment approach

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Obtain potential outcome estimates

$$\hat{Y}_i^{(0)} = \hat{m}(X_i, T_i = 0)$$

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Estimate average treatment effects

$$\hat{\tau} = \frac{1}{n} \sum_{i=1}^n \hat{Y}_i^{(1)} - \hat{Y}_i^{(0)}$$

Let's see some code

IPW vs G-computation

Both estimate average treatment effect but take two different approaches

IPW

Models the treatment

$$\hat{q}(X_i) = \hat{\mathbb{P}}(T_i = 1 | X_i)$$

Use IPW when you believe your treatment model is accurate

G-computation

Models the outcome

$$\hat{Y} = \hat{m}(X, T)$$

Use G-computation when you believe your outcome model is accurate

IPW vs G-computation

Both estimate average treatment effect but take two different approaches

IPW

Models the treatment

$$\hat{q}(X_i) = \hat{\mathbb{P}}(T_i = 1 | X_i)$$

Use IPW when you believe your treatment model is accurate

G-computation

Models the outcome

$$\hat{Y} = \hat{m}(X, T)$$

Use G-computation when you believe your outcome model is accurate

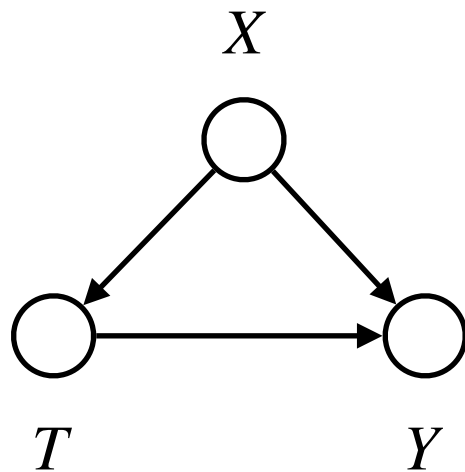
Augmented IPW: combines IPW & G-computation

Requires either treatment model or outcome model to be accurate

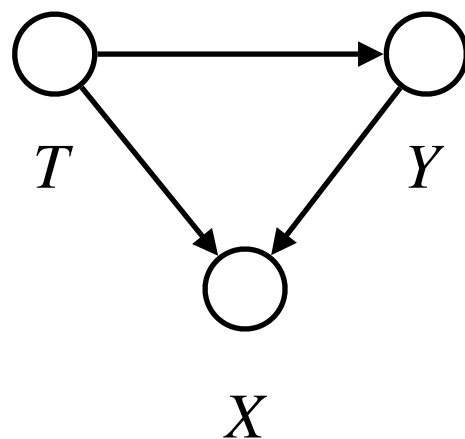
**How do we know what variables
to include in our model?**

Graphs are useful for this!

How do we know what variables to include in our model



X is an observed confounder —
include X (Simpson's paradox)



X is downstream of T and Y —
do **not** include X

Sometimes called a *collider*
variable (Berkenson's paradox)

Causal Inference and Statistics

Important - *causal inference rests on untestable assumptions*

Example [D'Agostino McGowan et al. 2023]: The statistical summaries and visualizations for the following datasets are identical, but the true causal effect differs

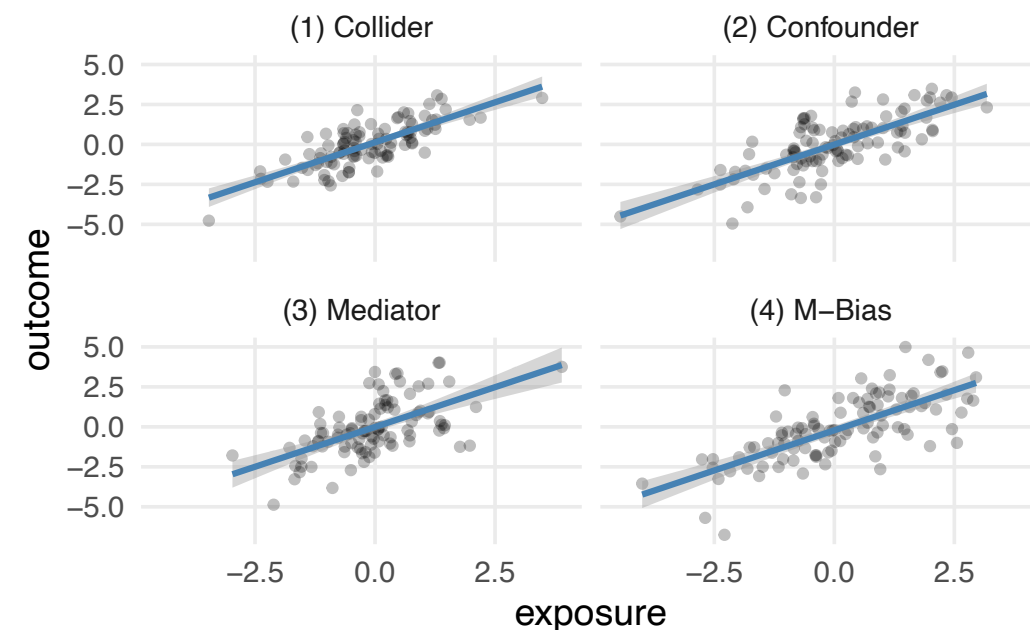
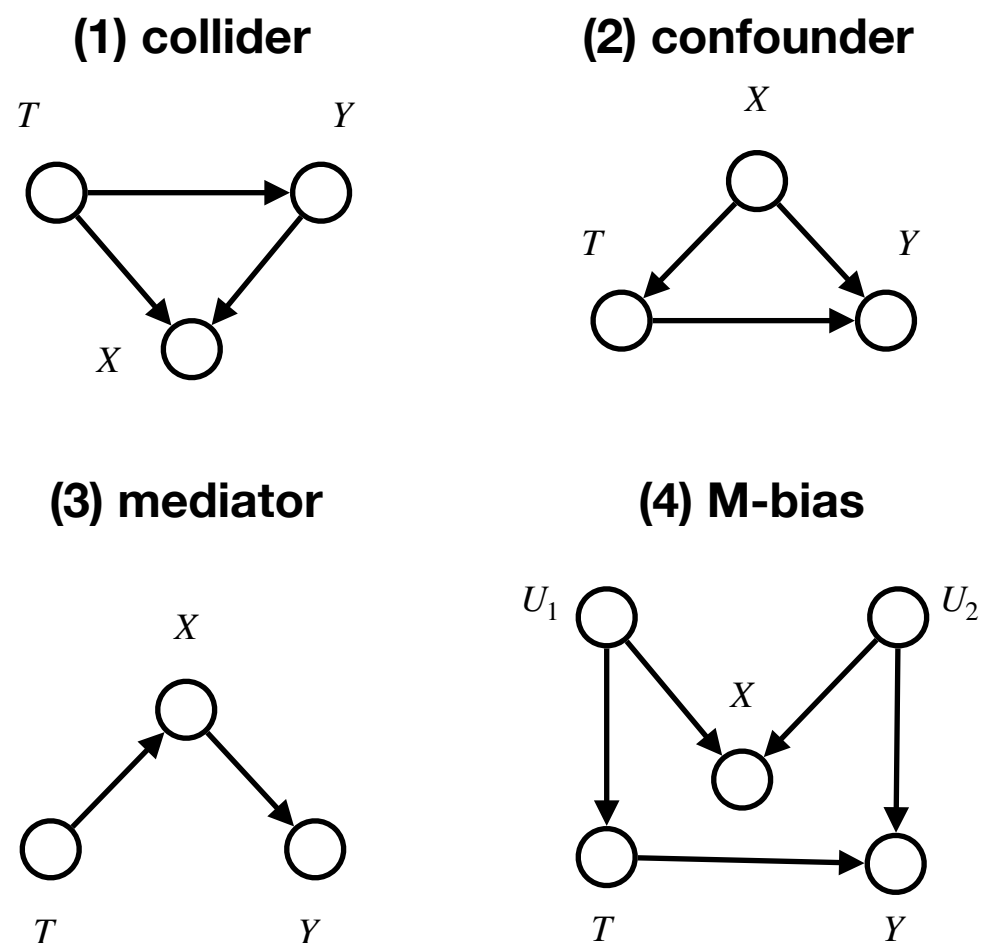
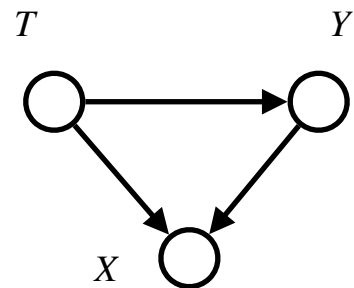


Figure 3: 100 points generated using the data generating mechanisms specified (1) Collider (2) Confounder (3) Mediator (4) M-Bias. The blue line displays a linear regression fit estimating the relationship between T and Y ; in each case, the slope is 1.

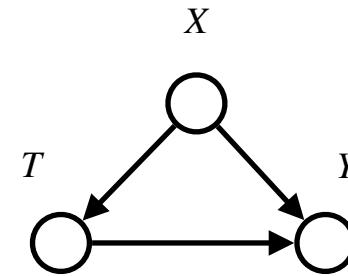
Causal inference and Statistics

(1) collider



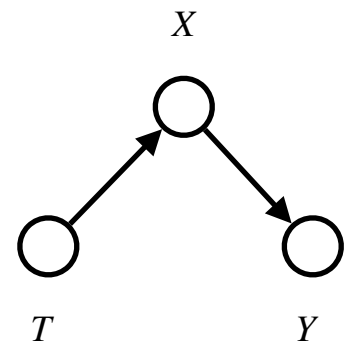
$$\begin{aligned}
 T_i &\sim N(0, 1) \\
 Y_i &= T_i + \varepsilon_i^y, \quad \varepsilon_i^y \sim N(0, 1) \\
 X_i &= 0.45T_i + 0.77Y_i + \varepsilon_i^z, \quad \varepsilon_i^z \sim N(0, 1)
 \end{aligned}$$

(2) confounder



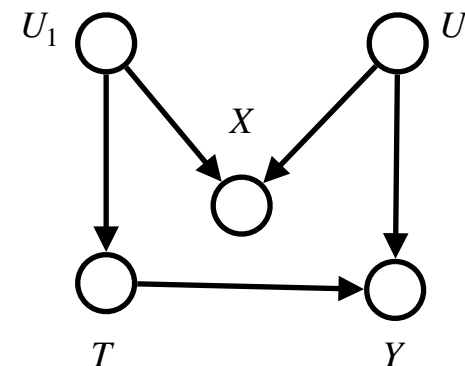
$$\begin{aligned}
 X_i &\sim N(0, 1) \\
 T_i &= X_i + \varepsilon_i^t, \quad \varepsilon_i^t \sim N(0, 1) \\
 Y_i &= 0.5T_i + X_i + \varepsilon_i^y, \quad \varepsilon_i^y \sim N(0, 1)
 \end{aligned}$$

(3) mediator



$$\begin{aligned}
 T_i &\sim N(0, 1) \\
 X_i &= T_i + \varepsilon_i^x, \quad \varepsilon_i^x \sim N(0, 1) \\
 Y_i &= X_i + \varepsilon_i^y, \quad \varepsilon_i^y \sim N(0, 1)
 \end{aligned}$$

(4) M-bias



$$\begin{aligned}
 U_{1i} &\sim N(0, 1) \\
 U_{2i} &\sim N(0, 1) \\
 X_i &= 8U_{1i} + U_{2i} + \varepsilon_i^x, \quad \varepsilon_i^x \sim N(0, 1) \\
 T_i &= U_{1i} + \varepsilon_i^t, \quad \varepsilon_i^t \sim N(0, 1) \\
 Y_i &= T_i + U_{2i} + \varepsilon_i^y, \quad \varepsilon_i^y \sim N(0, 1)
 \end{aligned}$$

Data generating mechanism	$\hat{ATE}$		Correlation of X and T	Correct causal effect
	not adjusting for X	adjusting for X		
(1) Collider	1	0.55	0.7	1
(2) Confounder	1	0.50	0.7	0.5
(3) Mediator	1	0.00	0.7	Direct: 0, Total: 1
(4) M-Bias	1	0.88	0.7	1

Let's see some code

Solutions

- **Key Assumption:** *no unmeasured confounders* (but we have observed confounders!)

Matching

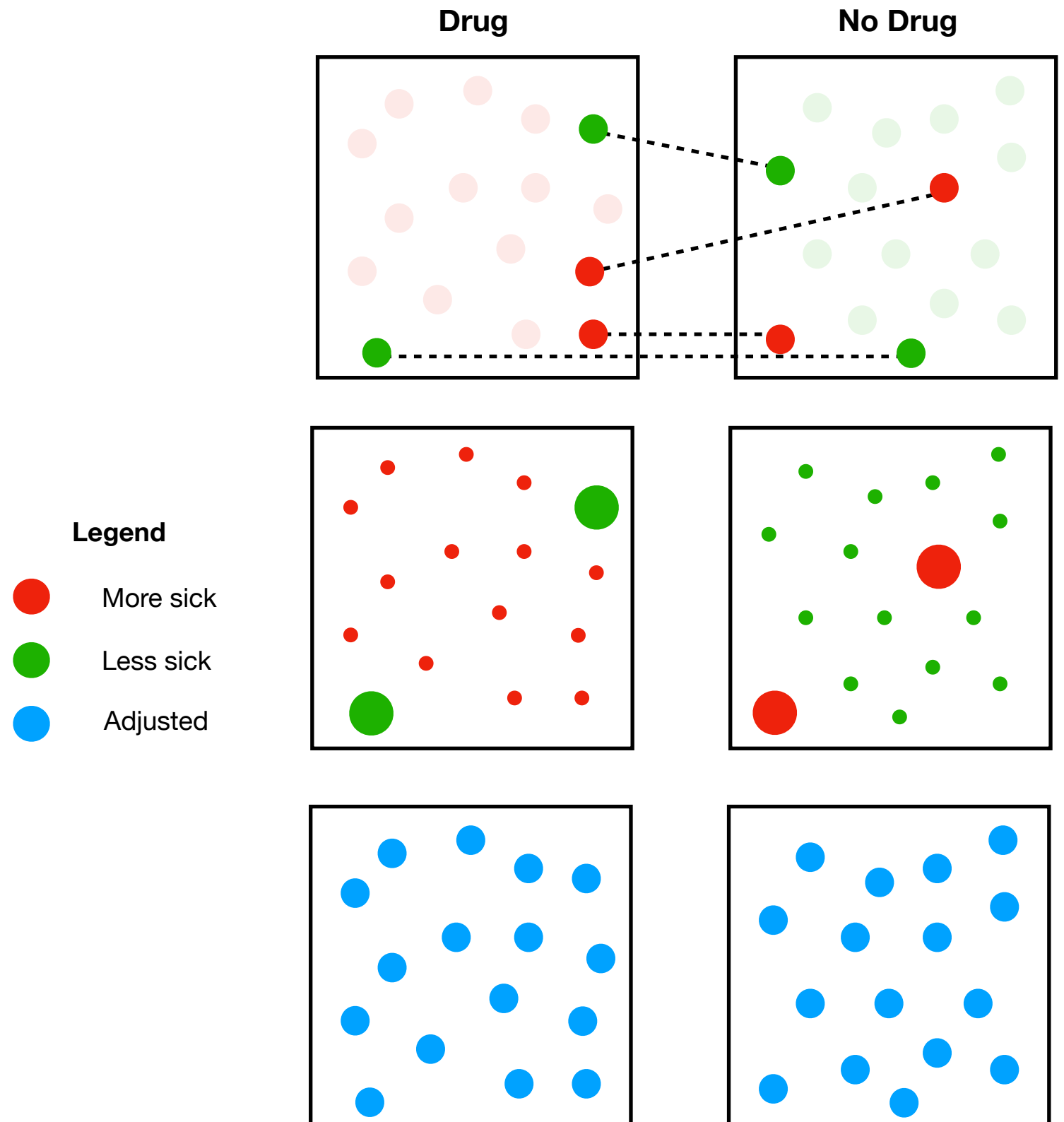
- Pair datapoints from each group that have same characteristics
- Calculate drug effect using difference between pairs

Propensity Score Reweighting

- Assume a model for whether drug is assigned. Re-weight data so patient characteristics are more balanced when calculating the drug effect

Adjustment (G-computation)

- Assume a model for how sickness affects lifespan, and adjust for sickness when calculating the drug effect

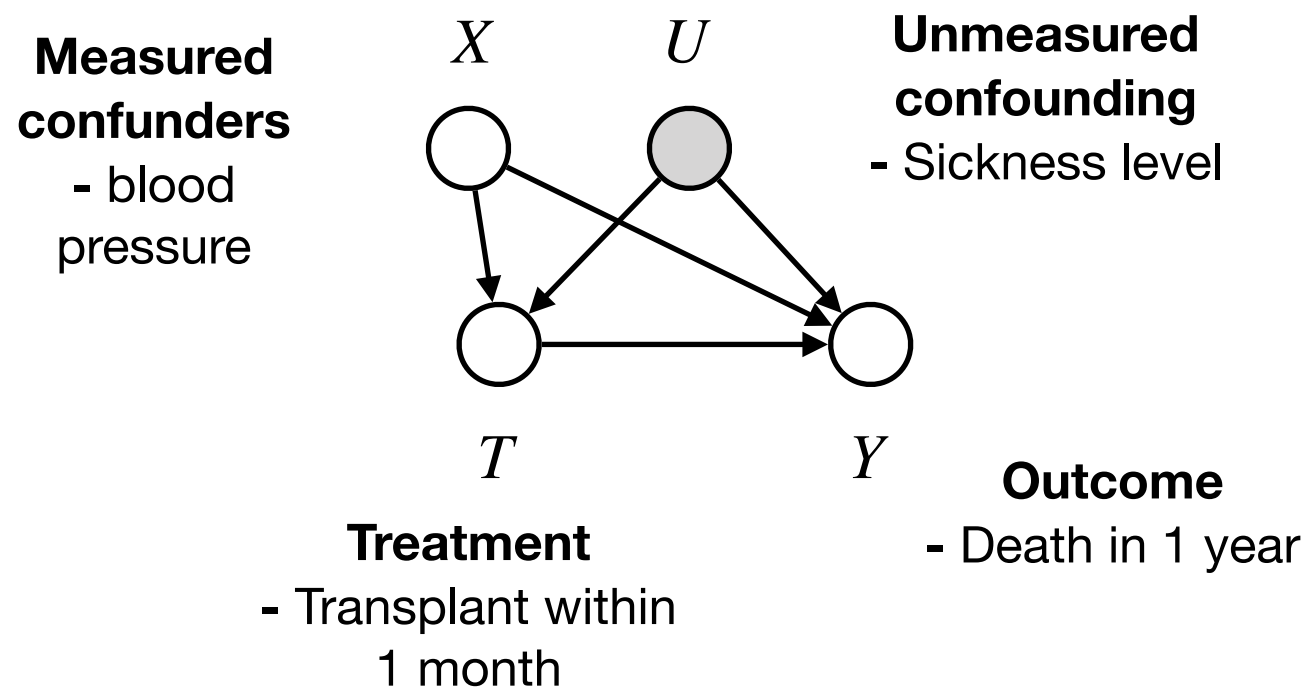


So far: all confounders are observed

i.e. we can find some covariates X that remove confounding between Y and T

$$Y_i^{(0)}, Y_i^{(1)} \perp T_i \mid X_i.$$

But this assumption may not be satisfied



- **Key Assumption:** *no unmeasured confounders* (but we have observed confounders!)

What can we do if we think there are unobserved confounders?

1. Sensitivity analysis:

- Assume there is an unmeasured confounder that changes the propensity score (i.e. a confounder makes it more or less likely to get treatment)
- See how sensitive the estimated treatment effect is to this change - would the unmeasured confounder need to be very large to change the conclusion?

2. Utilize natural experiments (briefly, next)

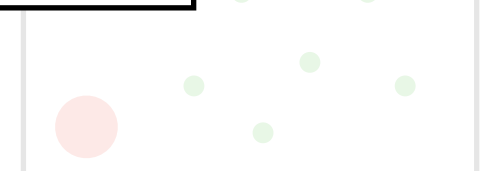
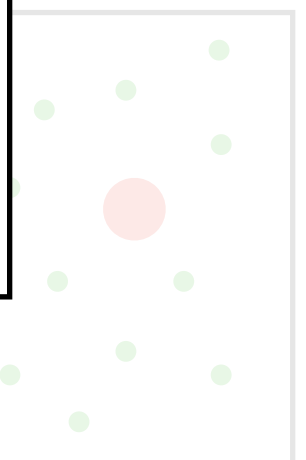
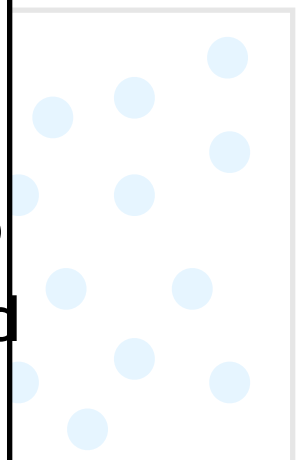
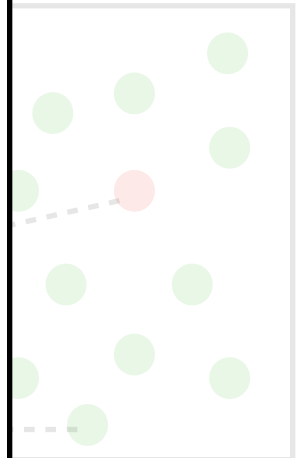
- Pair datapoints that have same
- Calculate drug difference bet

- Assume a model affects lifespan sickness when drug effect

- Re-weight data characteristics when calculating the drug effect

Drug

No Drug



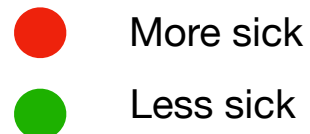
Solutions

- **Key Idea:** deal with *unobserved confounding* using natural experiments

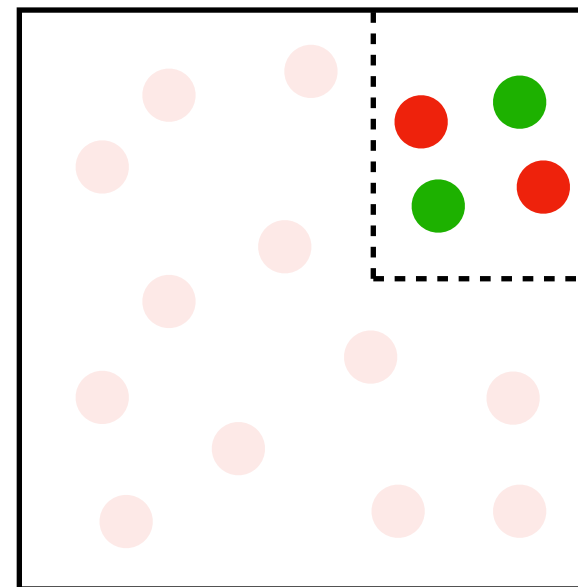
Focus on “like random” subset

- This includes methods such as instrumental variables, regression discontinuity studies

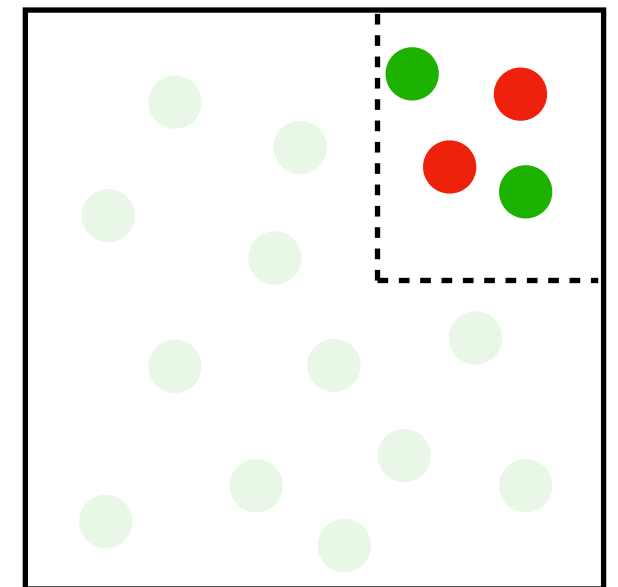
Legend



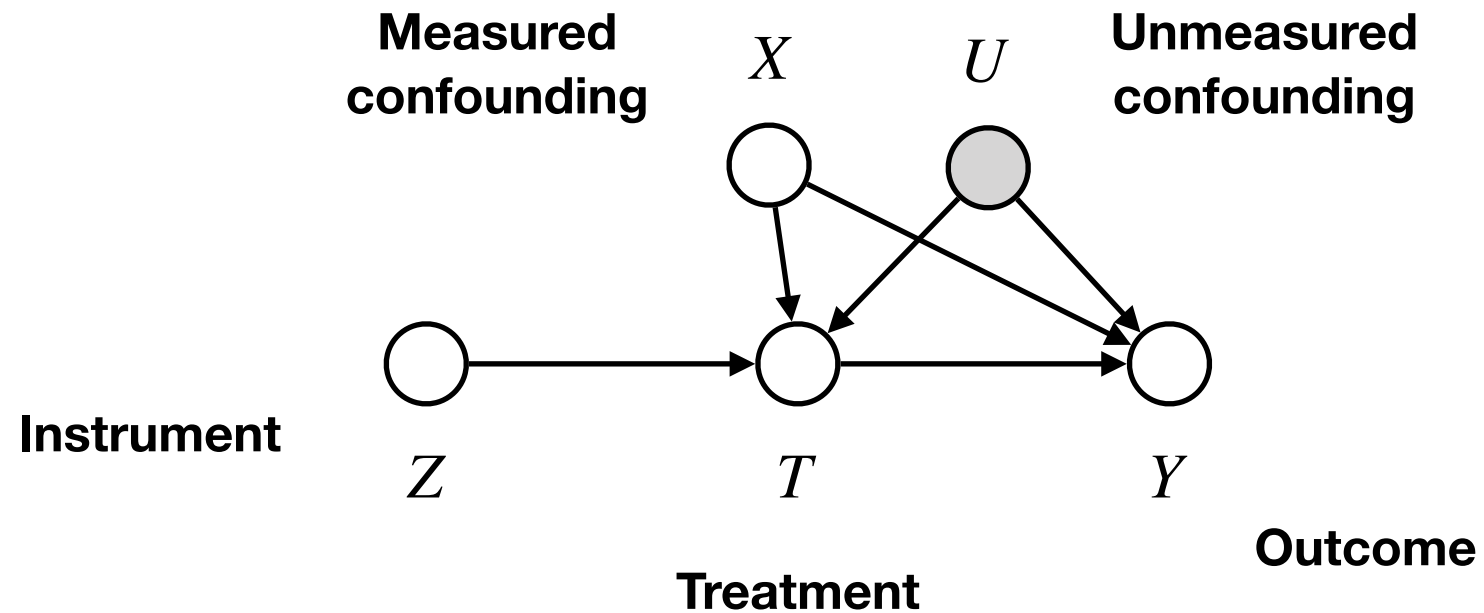
Drug



No Drug



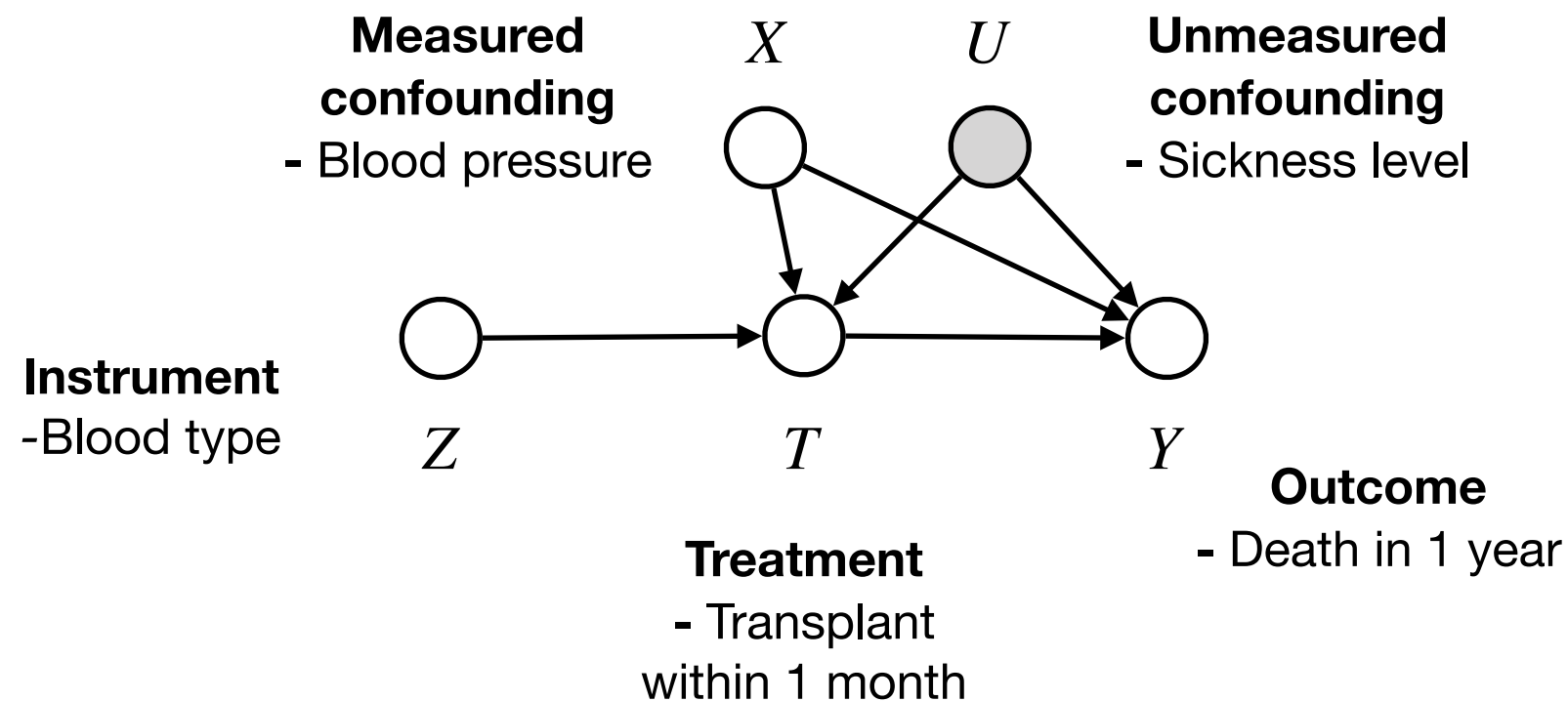
Instrumental Variables



Idea: if we can't randomize treatment assignment, how can we leverage randomness in the world to identify the causal effect?

The inventors of this idea received a Nobel Prize in 2021

Example: Effect of Organ Transplant Timing



- Patients who get transplants quickly are both (i) more sick and (ii) more likely to have poor survival outcome due to that sickness level — that is, sickness level is an **unmeasured confounder**
- **Instrument:** patients with blood type AB can accept any blood type, while O-type patients have to wait for an O-type transplant. Blood type is a proxy for random treatment assignment.

Instrumental Variables

Two regressions:

1. Regress T as outcome on Z as covariate

$$\hat{T} = \hat{m}_F(Z, X)$$

$$\text{e.g. } \hat{T} = \hat{\beta}_1 Z + \hat{\beta}_2 X$$

2. Regress Y on $\hat{T}$ and X

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Compute average treatment effect

$$\hat{Y}_i^{(0)} = \hat{m}_S(\hat{T}_i = 0, X_i)$$

$$\hat{Y}_i^{(1)} = \hat{m}_S(\hat{T}_i = 1, X_i)$$

$$\hat{\tau} = \frac{1}{n} \sum_{i=1}^n \hat{Y}_i^{(1)} - \hat{Y}_i^{(0)}$$

Instrumental Variables

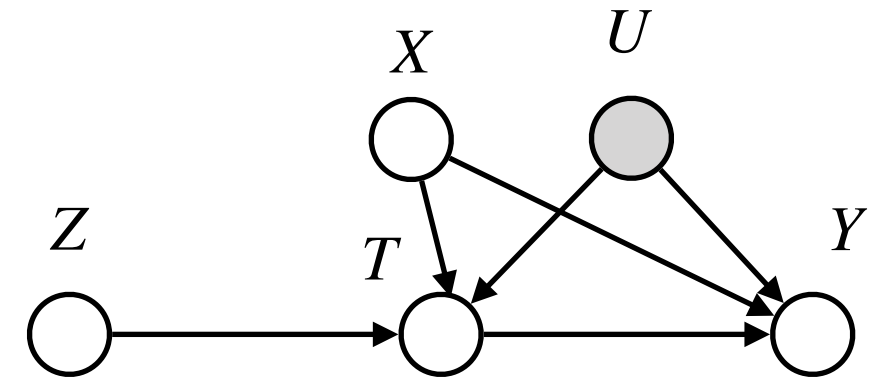
- **Assumptions:**

1. No unmeasured confounding in Z

2. Relevance: Z is not independent of T conditioning on X

3. Exclusion restriction: Z only affects Y through T

- Generally, these assumptions need to be evaluated based on the application or domain



Instrumental Variable Examples

Outcome variable	Treatment variable	Source of instrumental variable (natural experiment)	Reference
Mortality	Delivery of premature baby in high level NICU vs. regular hospital	Mother's differential travel time from nearest high level NICU to nearest regular hospital	Baiocchi, Small, Lorch and Rosenbaum (2010)
Earnings	Years of schooling	Proximity to college when growing up	Card (1995)
Earnings	Years of schooling	Quarter of birth	Angrist and Krueger (1991)
Achievement test scores	Class size	Discontinuities in class size due to maximum class-size rule	Angrist and Lavy (1999)
Health	Heart attack surgery	Proximity to cardiac care centers	McClellan <i>et al.</i> (1994)
Crime	Police	Electoral cycles	Levitt (1997)
Employment and earnings	Length of prison sentence	Randomly assigned federal judge	Kling (1999)
Birth Weight	Maternal smoking	State cigarette taxes	Evans and Ringel (1999)

Summary

- **Key Assumption:** *no unmeasured confounders* (but we have observed confounders!)

Matching

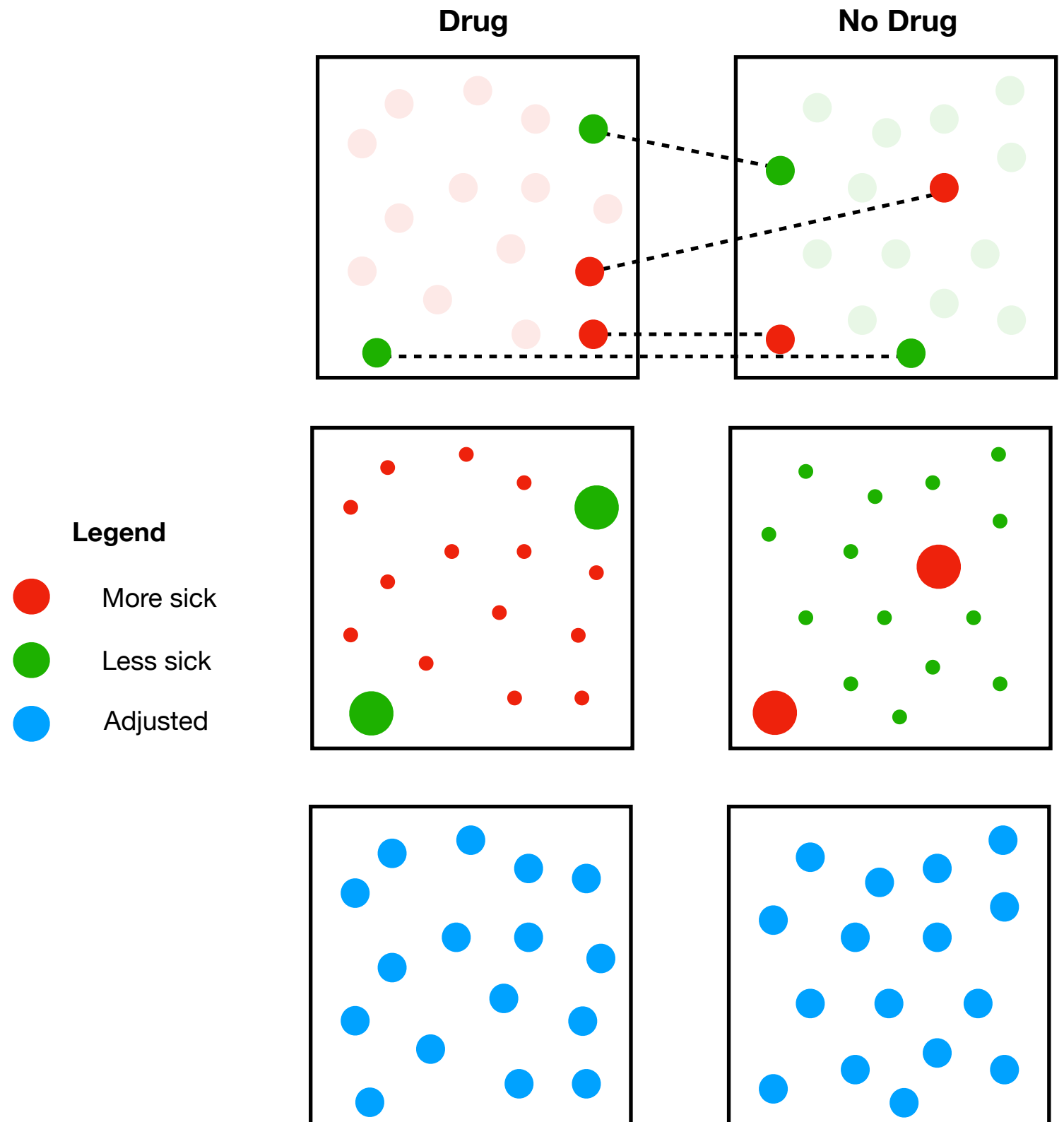
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Summary

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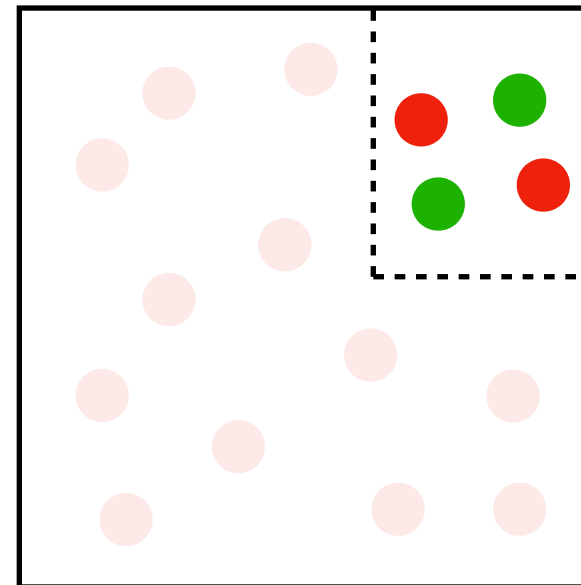
Focus on “like random” subset

- This includes methods such as instrumental variables,

Legend

- More sick
- Less sick

Drug



No Drug

